

BLOCK 3
EXTENSION IN RURAL DEVELOPMENT

BLOCK 3 EXTENSION IN RURAL DEVELOPMENT

Extension is regarded as the most effective strategy for rural development in developing countries in order to meet the major challenges of rural development are poverty, illiteracy, unemployment, homelessness , etc. In this block an attempt has been made to introduce basic concept, philosophy, principle, approaches and methods of extension. The foundation of diffusion and adoption has been elaborated along with innovations and experiences in rural extension.

Unit 9 Concepts and Philosophy of Extension in Rural Development

The unit aims at introducing you the concepts and philosophy of extension particularly in rural development. Extension aims at bringing positive changes in the behaviour of human being where communication acts as a catalyst facilitating exchange of information and ideas.

Unit 10 Approaches and Methods of Extension in Rural Development

Extension approaches have emerged in response to the diverse socio-political situations in different countries. Generalized extension, Training and visit extension, Commodity approach and project approach evolved out of the needs. There is ray of hope for *sustainability of extension system in participatory Agricultural* Extension Approach. A number of extension methods are being used. Group methods are most suitable and popular in general. However, mass media are gaining prominence. New communication technologies have helped in reaching farmers useful messages at the doorsteps.

Unit 11 Diffusion and Adoption of New Technologies

The units focus on diffusion process and its elements, attributes of innovations, stages of adoption, innovation decision process, adoption categories and its characteristics. It will throw light on rate of adoption and factors affecting rate of adoption.

Unit 12 Rural Extension: Innovations and Experience

The concept of rural extension related to rural development have been introduced. Critical elements, of different rural developmental programmes have been discussed to meet the challenges for attaining broader goal of increasing farm income and improving rural livelihoods.

UNIT 9 CONCEPTS AND PHILOSOPHY OF EXTENSION

Structure

- 9.0 AIMS and Objectives
- 9.1 Introduction
- 9.2 Rural Development
- 9.3 Rural Development Roadblocks
- 9.4 Concept of Extension
 - 9.4.1 Worldwide Experience
 - 9.4.2 Indian Experience
- 9.5 Disciplines of Extension Education
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- 9.7 Dimensions of Extension
- 9.8 Philosophy of Extension
- 9.9 Objectives of Extension
- 9.10 Principles of Extension
- 9.11 Let Us Sum Up
- 9.12 Keywords
- 9.13 Suggested Readings
- 9.14 Answers to Check Your Progress

9.0 AIMS AND OBJECTIVES

The Unit aims at introducing you the concepts and philosophy of extension. You will get conceptual clarity on extension activities in general and in rural development in particular. You will be familiar with the definitions of extension put forth by experts of various shades along with its general and specific objectives.

After going through this unit, you will be able to:

- Write various definitions of extension given by experts of various fields;
- Contextualize the concept of extension;
- Know the process and dimensions of extension;
- Be aware of extension activities practiced especially for rural development;
- Know the philosophy of extension;
- Figure out the objectives and principles of extension.

9.1 INTRODUCTION

Extension is heralded as the most effective strategy for rural development in poor countries. It aims at bringing a change in human behaviour where communication acts as a catalyst facilitating exchange of information and ideas. Esminger (1957) spells out the purpose of extension education as to change attitudes and practices of target population. Dahama (1973) defines extension education as a process to impart knowledge and skills to the rural people about the innovative practices in a manner they can adopt it in their local settings. Bhatnagar and Desai (1987) broadened it and defined it as to bring the desirable changes in the behaviour — knowledge, skills, attitude, understanding, goals and confidence — of people through mutual learning. The concept of extension that started as out of school system education has undergone sea change in its aims, objectives and target population. It is regarded as a process, a method, a system and an art for bringing out change in human behaviour. One of the important objectives of extension is to bring about changes in the practices of the rural people for increased production and income leading to the improvement in their standard of living. In this unit, we shall discuss the meaning, definition, concept, objectives and philosophy of extension.

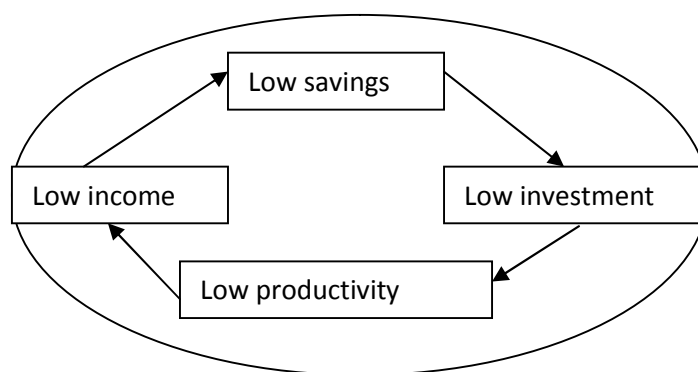
9.2 RURAL DEVELOPMENT

Rural development can be defined as a planned intervention to the socio-economic dimensions to achieve better living standard on a sustained basis of low-income population residing in rural areas. As a Phenomenon, it is the outcome of various physical, environmental, technological, economic, socio-cultural and institutional factors in the rural areas. As a Strategy, it is the design to cause the intended changes in the socio-economic and cultural life of the rural folks. As a discipline, it is multi-disciplinary; coming together of agricultural, social, behavioural, engineering and management sciences bringing together both agricultural and non-agricultural aspects of life.

9.3 RURAL DEVELOPMENT ROADBLOCKS

The major challenges to rural development are: poverty, illiteracy, unemployment, homelessness and crime. There is a vicious circle of poverty that has engulfed rural folks trapping them in a perennial state of it. Vicious Circle of Poverty can be defined as a process where the circular forces acting and reacting on one another in such a way as to keep a poor person or country in a state of perennial poverty. Poor countries have low income. Low income leads to low savings and investment. Low investment gives low productivity again leading to low income. Illiteracy makes one unable to identify, interpret, understand, create, communicate, and compute, text materials which are beneficial to them. Unemployment worsens their economic conditions so as homelessness. Crime in rural areas especially to women

when go unaddressed defeats the idea of a progressive society where women empowerment is one of its strong pillars.



(Vicious Circle of Poverty)

Check Your Progress 1

- Note:** i) Write your answer at the space provided.
ii) Check your answers with those given at the end of the unit.

1) Write a short note on the Vicious Circle of Poverty.

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9.4 CONCEPT OF EXTENSION

$$\text{Extension} = \text{Ex} + \text{Tensio}$$

Ex = out, *Tensio* (Latin roots) = Stretching

Extension = Stretching out/ reaching out

As given in the above equation, ‘extension’ is originated from the Latin roots, ‘ex’ meaning ‘out’ and ‘tensio’ meaning ‘stretching’. Literally meaning ‘stretching out’. For all practical purposes, it means reaching out to people, who otherwise can’t or shall not or seldom come, beyond the boundaries of educational and research institutes to educate them on various issues of importance. If it is used for rural people for their development, it is called extension for rural development.

9.4.1 Worldwide Experience

It is documented that the term 'extension' was first used in Britain in 1840s. James Stuart, Fellow of Trinity College, Cambridge is considered the father of extension for his several lectures to women's associations and working man's clubs in north of England during 1867-68. In 1873, Cambridge University; in 1876, London University and in 1878, Oxford University used it formally to take the educational advantages of universities to the doorsteps and workplaces of ordinary people. In 1880, his work became popular as Extension Movement.

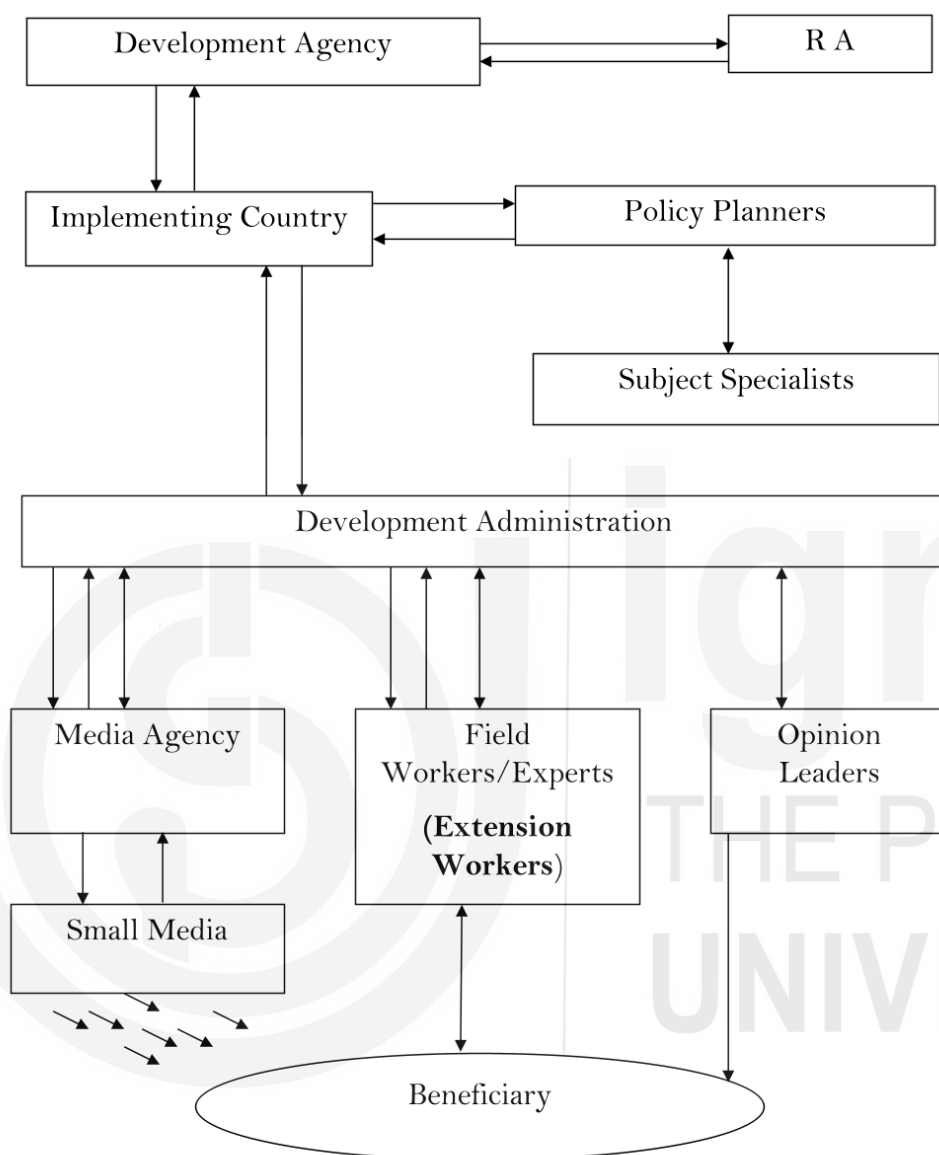
Towards the second half of 19th century, the word Extension was used for Extension education in the U.S.A as urbanisation led to migration of rural people to cities hitting the agriculture hard, and taking a toll on urban resources along with posing problems in education, employment and housing in urban areas. To reduce migration of rural people to urban areas and to engage rural people in agricultural and rural development programs President Roosevelt of the USA established a "county commission" in 1862 under the chairmanship of Liverty Hydevelle. The Commission observed that there is a vast difference between the standard of living between people living in rural and urban areas which is to be bridged. The rural people who need to uplift their standard of living can do so permanently only if the initiative comes from themselves. For this to happen it is essential to bring about changes in the field of knowledge, understanding, skill and attitude. The commission recommended to roll out educational work by establishing an institution that can organise out of school of education. Subsequently, Smith Leaver Act (1914) was passed leading to the integration of cooperative extension services and out of school education was organized. As an outcome, rural people were able to taste development in farming and housing. A new type of education emerged in practice and came to be known as Agricultural Extension Education.

9.4.2 Indian Experience

In India, the term Extension became popular with the establishment of National Extension Service in 1953. Since then, extension has been considered as the means for the overall development of the rural people. The National Commission on Agriculture, 1976 defines extension as an informal out of school education and services for the members of the rural family and others engaged directly or indirectly in agricultural production to facilitate and enable them to use in a sustained manner improved practices of production, management, conservation and marketing. Extension also aims at empowering people to such a point so that the beneficiary will seek means of improving one's farm occupation, home and family life in a holistic manner. Those who are tasked to carry out the extension activities are called extension workers. In the Indian context a few examples can be: Accredited Social Health Activists (ASHA) workers, Auxiliary Nurse Midwife (ANM)

workers, *Aanganwadi* workers educating the villagers on issues of health, mother and child care, education, family planning, agriculture among others.

A hypothetical extension campaign using extension workers is given below for better understanding of extension education.



(Fig.: A Hypothetical Extension Campaign. TD= Top-Down communication, BU= Bottom-Up communication, RA= Research Agency, H= Horizontal communication)

Suppose an international development agency get to know about a problem in some of part of the world from primary or secondary sources. Seeing the gravity of the issue, it decides to intervene. It may approach a Research Agency (RA) to further investigate and fact find the problems in details. That may involve both top-down and bottom-up communication. Once it gets the report from the RA and it is certain to intervene, it may approach the concerned country (say a Developing Country) for approval involving both top-down and bottom-up communication. The developing country may approach its policy planners for the need of such intervention and priority involving top-down and bottom-up communication. The policy planners, in

turn, may approach subject specialists for expert opinion involving horizontal communication. Once, the country gets green signal from the policy planners, it can permit the international agency for carrying out its work in the country.

The international agency may approach the district administration through various central planning authorities with top-down and bottom-up communication. Now, development administration or district administration can prepare for its execution at the grassroots. It may hire, in consultation with the international development agency, extension workers/field workers, a media agency and opinion leaders for the purpose and involve in top-down, bottom-up and horizontal communication.

The required number of extension workers can be trained by experts of both the international development agency and district administration. The district administration may further facilitate for carrying out the extension activities in the targeted place to the target population or beneficiary with horizontal communication.

The media agency hired may further employ small media involving top-down and bottom-up communication like folk media, community media, video strips, and hoardings among others to create a conducive environment for the beneficiary to adopt the innovation propagated by the extension workers.

To defeat the misunderstandings, rumours and deal with hesitancy to adoption of innovation disseminated by the extension workers, opinion leaders may be pressed into action involving horizontal communication by district administration and international development agency. The opinion leaders may be involved with the beneficiary with top-down and bottom-up communication.

Any feedback from the beneficiary may be collected by extension workers, opinion leaders and to some extent by small media practitioners that can be passed on to the district administration/development agency which can be further passed on upwards. At the appropriate level course correction can be done to restart the process again.

9.5 DISCIPLINES OF EXTENSION EDUCATION

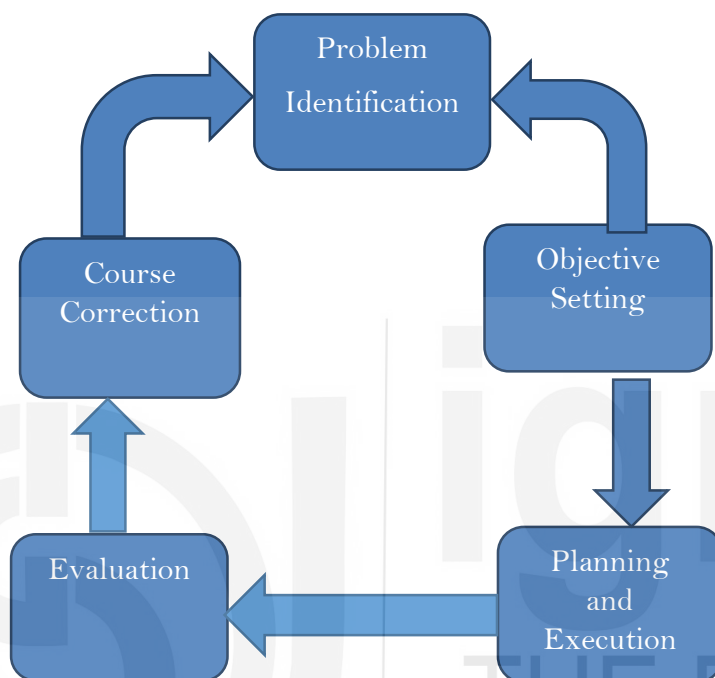
Depending upon the application of extension in various fields of need, the disciplines of extension can be found in Agricultural, Dairy, Fishery, Health, Sanitation, Micro Finance, Agricultural Engineering, Home Science, Fruit & Vegetable, Forestry, Sericulture, Veterinary, Medical and the list goes on.

9.6 PROCESS OF EXTENSION EDUCATION

A process can be defined as a systematic series of actions directed to some end. It involves some steps. This is on-going and continuous. J.P. Leagans

(1961) talks about five essential and interrelated steps for an effective extension. These steps can be renamed as:

- Problem Identification
- Objective Setting
- Planning and Execution
- Evaluation
- Course Correction



(Fig.: Five Steps of Extension Process)

Problem Identification

As the name suggests, this step consists of identifying the problems after collection and careful analysis of data — the beneficiary's work, their socio-economic, physical and technological environment — which can be assembled by secondary literature and appropriate field work.

Objective Setting

After identifying the problems, it is time to set achievable targets by the beneficiary which are clearly spelt out can bring positive changes and the top priority areas of concern.

Planning and Execution

Proper planning should be there before executing the campaign. It is about What and How. What to teach and How to teach. The content and the method of teaching should be planned in minutest of details and accordingly steps need to be taken to execute in the field.

Evaluation

Post execution, this step is essential to find out the efficacy of the campaign. Whether the work is going on as per plan and bringing intended results or not.

Course Correction

The lacunae of the campaign identified during the step of Evaluation can be considered for corrections in our problem identification, objective setting and overall planning and execution. Thus, the process goes on in a continuous manner.

Check Your Progress 2

Note: a) Write your answer at the space provided.

b) Check your answers with those given at the end of the unit.

1) Write a short note on various steps involved in the process of extension.

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9.7 DIMENSIONS OF EXTENSION

An effort was made by D.C. Misra in 1990 to spell out the dimensions of extension. He analysed various dimensions from 1953 to 1989 and carved out 10 dimensions of extension as given below:

- 1) Altruistic
- 2) Educational
- 3) Communication
- 4) Behavioural
- 5) Technology
- 6) Research
- 7) Inputs
- 8) Professional
- 9) Income
- 10) Management

Altruistic

Altruism is all about unselfish concern for the welfare of others. Extension believes in being philanthropic and altruistic. The approach of extension should have altruism. None should be left out. It should be inclusive, making all possible efforts to reach the unreached.

Educational

Education brings intended changes in one's behaviour whether through formal, informal education or non-formal education. Primarily extension education is non-formal in type. Formal education has a predefined beginning and end; informal education has only the beginning but non-formal education has a definite beginning and end with lots of flexibilities. These flexibilities make one, otherwise left out, to learn. Goal of educational dimension is to bring changes in knowledge, skill and attitude of people with whatever method appropriate for behavioural changes.

Communication

Communication in extension aims at being horizontal. The sender of the message should come down to the level of the receiver's understanding and encode the message to avoid any difficulty in decoding by the receiver. Lesser the noise, better is the effectiveness of communication.

Behavioural Dimension

The foundation for all developmental initiatives is the change in behaviour. The constituents of behaviour, viz., knowledge, skill and attitude are to be addressed thus. Behavioural dimension gives emphasis on favourable changes in knowledge, skill and attitude.

Technology

In Indian context this dimension largely means transfer of technology to the needy people. As skills make technology operational, necessary skills have to be imparted for this purpose along with taking feedback on the technology to its inventors for possible modifications.

Research

The research dimension has two aspects. One, that produces technology which are to be transferred to the needy and second, on extension itself that provides methodological support for carrying out extension services and contributing for development of it as an educational discipline.

Inputs

With change in technology, new inputs are essential. New technologies are more and more input driven. Different types of inputs are needed to kick start production process. Whether tangible (physical products) and intangible (information) inputs, it becomes the responsibility of the agency for its timely availability. This can bring big results in our extension campaigns.

Professional

The discipline of extension is a well-established profession as it has its own syllabus to study, areas to do research on and academic programmes to teach. However, over a period of time the concept of extension has been understood as giving everything for free by some agency keeping educational aspect to it as secondary. However, agencies have their limitations. For how long they

would provide items free of cost is a debatable topic. May be the concept of sharing the cost of extension be deliberated upon.

Income

There should be increase in income of the beneficiary of extension. This can be most likely achieved when they adopt innovations. Adoption of innovations involve risks. Sometimes, it might not happen. In this case, there is something wrong somewhere. And the strategy needs to be revisited.

Management

To avoid failure of extension activities which are well intended management aspects of extensions are worth considering. Management of extension can be defined as the process by which human resources (people), technology, job in hand and other resources are joined in a well-coordinated manner to achieve extension objectives. The concepts of single line of command, reasonable span of management, professionalization of extension service, institutionalization of Farmer-Extension-Research linkage, monitoring, evaluation among others can be incorporated.

9.8 PHILOSOPHY OF EXTENSION

Philosophy is derived from two Greek words 'Philos' and 'Sophia'. *Philos* means knowledge and *Sophia* means manner lending the definition of Philosophy as the manner of achieving knowledge with a particular system of thought. Philosophy of extension aims at a person to effectone's self development and that of community through one's leadership by adopting scientific approach and democratic means. Its core principle is rational self-interest e.g., the interest of the community must not be a victim because of personal interest and considers the development of an individual is crucial for the welfare of a family, community, society and the country.

Prof. Rudramurty (1964) relates the philosophy of extension education to philosophical views of the Vedas, the Upanishads and the Bhagwad Gita that are necessary for human existence. Kelsey and Hearne (1 967) put the basic philosophy of extension as to teach people how to think, not what to think.

D. Ensminger describes the philosophy of Extension the given way:

- 1) Aims at changing the knowledge, attitudes, understanding and skill of the beneficiary. It is an educational process.
- 2) Is working with the beneficiary to answer their needs and wants ways to satisfy those.
- 3) Is "helping people to help themselves."
- 4) Is working by social learning i.e., learning by observation.
- 5) Holistic development by developing individuals and their world.
- 6) Working together common welfare and happiness of the people.

- 7) Working in tune with the culture of the beneficiary.
- 8) Involves a two-way process of communication.
- 9) Is a continuous educational process.

Prof. O.P. Dahama observes the following aspects of philosophy of extension education:

- Self-help is key.
- Considers people as greatest resources.
- A cooperative exercise.
- Rooted in democracy.
- Two-way sharing of knowledge and experience.
- Based on self-learning.
- Beneficiary participation is voluntary and cooperative.
- Persuasion and education of beneficiary.
- Customised programme based on the attitudes and values of the beneficiary.
- It is a never-ending, continuous process.

9.9 OBJECTIVES OF EXTENSION

The broad objective of extension is to raise the standard of living of rural folks by making them participate in decision making of optimal use of their natural resources. They are given a helping hand in increasing the agricultural production and income; improving the village arts and crafts leading to increased income; and improving the family life and living conditions by intervening in their sanitation, health, education and housing facilities if required.

According to National Commission on Agriculture (1976), agricultural extension aims at improving the efficiency of the human capital. Ban and Hawkins (1988) observes extension as a process that helps farmers analyse their present, anticipate future problems, increases existing knowledge and skills to face, motivates farmers to implement their choice and evaluate their opinion.

The specific objectives are:

- To help people in the exploration and analysis of their problems and needs.
- To groom organic leadership to help solve local problems.
- To spread specialized technical information in understandable terms which people can use and be benefitted out of it.
- To help in resource mobilisation.

- To collect feedback for solving management problems.
- To groom confident, independent, constructive, capable, efficient rural citizens who love their motherland.

The general objectives of extension can also be grouped as given below:

- Knowledge Development
- Attitude Development
- Skill Development

Knowledge Development

P. Harrison (1986) suggests that knowledge should be imparted to developing skills in information, relating it, interpret it, generalize it, apply it for solving problems, and developing the desire for continuous learning.

Attitude Development

Attitude is the tendency or orientation of the mind. An appropriate attitude is crucial for the success and failure of any programme. An extension educator should keep in mind the beneficiary's past experiences, attitudes and values before, during and after the extension programme.

Skill Development

Skill learning is crucial as the right kind of actions by the beneficiary depends on it. Knowledge can be put to best use by skills. Along with skills for farm practices rural people need life skills and new skills to avail new opportunities thrown open by various agencies these days.

Check Your Progress 3

Note: a) Write your answer at the space provided.

b) Check your answers with those given at the end of the unit.

- 1) Write a short essay on the objectives of extension education.

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9.10 PRINCIPLES OF EXTENSION

Experiences in the field of extension services and rural development from various programmes from India and abroad have contributed to the various guidelines to guide future decisions and actions in extension activities. These principles are essential for the success of activities and can be mentioned as:

- Need and interest
- Cooperation and participation
- Cultural difference
- Adaptability in teaching
- Learning by doing
- Trained specialists
- Adaptability
- Leadership
- Whole family
- Satisfaction
- Evaluation
- Neutrality
- Encouragement

Need and Interest

Extension work is self-learning activity. To make it acceptable to people, it should be made need based and interesting. This would also extract cooperation from people.

Cooperation and Participation

Without people's cooperation and participation, the extension work cannot be successful. Participation of target population brings confidence in them. It is suggested to rope in as many persons as possible to take part in an extension campaign.

Cultural Difference

Rural people are deep rooted in their culture. Extension workers should do everything possible to avoid cultural clash with them to ensure smooth execution of the activity.

Applied Science and Democratic Approach

The aim of extension workers is certainly to diffuse scientific innovations and instil scientific approach in them but in a democratic way as democratic principles are the pillars of extension approach. A lot of discussions at the grassroots level can take place between the beneficiaries and extension workers. Which is healthy.

Learning By Doing

Extension education emphasises gaining practical knowledge and experience the real time problems to sort out then and there; which is possible by the principle of learning by doing.

Trained Specialists

Based on the demand of the subjects and sensitivity of issues that can't be dealt by the extension workers well; sometimes specialists should step in as extension workers.

Adaptability in Teaching

The extension workers should adapt to new teaching methods keeping in mind the demographic and geographic variables of people. Their age, gender, education, economic status among others plays a role in deciding the right method or mix of methods to be applied for teaching.

Leadership

The cooperation and if possible, participation of the local leaders in the extension work is crucial for its success. They work as opinion leaders and help forming a favourable opinion about the work in progress to thwart any resistance out of ignorance or vested interests.

Whole Family

The rate of success of extension work is very high if all the members of a family take part in the exercise. This paves the way for faster behavioural changes in the members of family.

Satisfaction

People get satisfaction only after getting appreciable results else lose interest. Priority should be given to those work where there is scope for immediate results leading to people's satisfaction.

Evaluation

As discussed before extension education needs periodic evaluation to find out the efficacy of the programme. The results of evaluation will determine the future course of action. It generates confidence in people.

Neutrality

Extension workers should try to remain neutral in various areas while dealing with the local population. It may be local politics, caste system, local traditions among others. The scope derailing the activity is high if neutrality is not maintained.

Encouragement

Encouragement and not pressure go a long way in determining the success of extension activities. Active workers should be sufficiently encouraged to participate and remain active in extension activities.

Check Your Progress 4

Note: a) Write your answer at the space provided.

b) Check your answers with those given at the end of the unit.

1) Write a short note on the principles of extension.

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9.11 LET US SUM UP

The unit aims at introducing you the concepts and philosophy of extension particularly in rural development. Extension is regarded as the most effective strategy for rural development in developing countries. The major challenges to rural development are poverty, illiteracy, unemployment, homelessness and crime. Extension aims at bringing positive changes in the behaviour of human being where communication acts as a catalyst facilitating exchange of information and ideas. Literally meaning ‘stretching out’ and for all practical purposes, it means reaching out to people, who otherwise can’t or shall not or seldom come, beyond the boundaries of educational and research institutes to educate them on various issues of importance e.g., health, mother and child care, education, family planning, agriculture among others. If it is used for rural people for their development, it is called extension for rural development.

Depending upon the application of extension in various fields of need, the disciplines of extension can be divided as Agricultural, Dairy, Fishery, Agricultural Engineering, Home Science, Fruit & Vegetable, Forestry, Sericulture, Veterinary, Health & Family Planning, Micro-Finance and the list goes on. Extension education, as per J.P. Leagans (1961) follows five essential and interrelated steps than can be called as Problem Identification, Objective Setting, Planning and Execution, Evaluation and Course Correction. Extension can have ten important dimensions such as Altruistic, Educational, Communication, Behavioural, Technology, Research, Inputs, Professional, Income and Management Dimension.

The philosophy of extension aims at a person to bring one’s self development and that of community through one’s leadership by scientific approach and democratic means. The objectives of extension are to help people in the exploration and analysis of their problems and needs; to groom organic leadership to help solve local problems; to spread specialized technical information in understandable terms which people can use and be benefitted out of it; to help in resource mobilisation both local and that they need from outside; to collect feedback for solving management problems; and to groom confident, independent, constructive, capable, efficient rural citizens who love their motherland. Experiences in the field of extension services and rural development have contributed to the formation of various guidelines to guide

future decisions and actions in extension activities. These principles are essential for the success of activities and can be termed as Need and Interest, Cooperation and Participation, Cultural Difference, Adaptability in Teaching, Learning by Doing, Trained Specialists, Adaptability, Leadership, Whole Family, Satisfaction, Evaluation, Neutrality and Encouragement.

9.12 KEYWORDS

Concept: An idea of something.

Top-down communication: Communication flowing from higher rank or authority to lower rank or authority.

Bottom-up communication: Communication flowing from lower rank or authority to higher rank or authority.

Horizontal communication: Communication flowing between same or similar rank or authority.

Dimension: The scope, importance of something.

Skill: A craft or ability where somebody has competence.

9.13 SUGGESTED READINGS

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Rutledge

9.14 ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress 1

- 1) Vicious Circle of Poverty can be defined as the circular forces acting and reacting on one another in such a way as to keep a poor person or country in a state of poverty. Poor countries have low income. Low income leads to low savings and investment. Low investment gives low productivity again leading to low income.

Check Your Progress 2

- 1) A process can be defined as a systematic series of actions directed to some end. Hence it involves some steps. The is on-going and continuous. Each initiative continues to evolve as the situation changes. J.P. Leagans, (1961) talks about five essential and interrelated steps for an effective extension. These steps can be renamed as:

- Problem Identification
- Objective Setting
- Planning and Execution
- Evaluation
- Course Correction

Check Your Progress 3

- 1) The specific objectives are:
 - To help people in the exploration and analysis of their problems and needs.
 - To groom organic leadership to help solve local problems.
 - To spread specialized technical information in understandable terms which people can use and be benefitted.

- To help in resource mobilisation both local and that they need from outside.
- To collect feedback for solving management problems.

Check Your Progress 4

Experiences in the field of extension services and rural development from various programmes from India and abroad have contributed to the various guidelines to guide future decisions and actions in extension activities. These principles are essential for the success of activities and can be mentioned as:

- Need and interest
- Cooperation and participation
- Cultural difference
- Adaptability in teaching
- Learning by doing
- Trained specialists
- Adaptability
- Leadership
- Whole family
- Satisfaction
- Evaluation
- Neutrality
- Encouragement

UNIT 10 APPROACHES AND METHODS OF EXTENSION

Structure

10.0 Aims and Objectives

10.1 Introduction

10.2 Extension Approaches

- 10.2.1 General Extension Approach
- 10.2.2 Commodity Specialized Approach
- 10.2.3 Project Approach
- 10.2.4 The Training and Visit Approach
- 10.2.5 Farming Systems Approach
- 10.2.6 Privatised Extension Approach
- 10.2.7 Participatory Agricultural Extension
- 10.2.8 Education Institution Approach
- 10.2.9 Market-led Extension Approach
- 10.2.10 Cyber extension approach

10.3 Methods of Extension

- 10.3.1 Individual Contact Methods
- 10.3.2 Group Extension Methods
- 10.3.3 Mass Methods
- 10.3.4 New Communication Technology

10.4 Let Us Sum Up

10.5 Keywords

10.6 Suggested Readings/References

10.7 Answers to Check Your Progress

10.0 AIMS AND OBJECTIVES

After studying this unit, you should be able to:

- state different types of approaches in extension;
- discuss need and significance of different types of extension approaches;
- explain characteristics of different extension approaches;
- describe reasons for choosing a particular extension approaches;
- discuss different types of extension methods;
- describe special characteristics of different types of methods; and
- explain features of different extension methods and media.

10.1 INTRODUCTION

Agricultural Extension as profession is social innovation of the nineteenth century. It originated in the wake of industrialization to take the educational

benefits to rural masses. The early approaches of extension work were mere voluntary efforts of enlightened few. Later extension work had changed approaches of working during past one century due to various experiences in different socio-political situations. Thus, we have multiple approaches to respond to different situations such as project approach, commodity approach, privatized, participatory and others. Similarly, many methods emerged out of experiences of working such as individual to group and mass method in this unit an attempt has been made to describe different approaches, methods and media.

10.2 EXTENSION APPROACHES

Extension work is done through many ways to bring about transformation in rural areas. Depending on the socio-economic and political situations, diverse approaches have been adopted to respond to particular needs. Depending on the nature of intervention, approaches of extension have been classified as below:

- Generalized Extension Approach
- Commodity Specialized Extension Approach
- Educational institution Extension approach
- Project Extension Approach
- Training and visit Extension Approach
- Farming systems development Extension approach
- Privatized Extension Approach
- Participatory Extension Approach.

10.2.1 General Extension Approach

This is basic approach of extension work which assumes that technologies appropriate for local people are available but not being used. If knowledge of these technologies can be communicated to farmers, farm practices would be improved. The purpose is to help farmers increase their production. In order to reach technologies to the people, government intervenes through structured extension services. Large number of field staff and hierarchy of middle and high level personnel are appointed. Arrangements are made to supply inputs, training, credit, advice, etc. Large number of field extension personnel are required. Resources required are also high, with central governments bearing most costs. Success is measured through the adoption rate of recommendations and increases in national production. This approach is considered as a top to bottom approach.

10.2.2 Commodity Extension Approach

Commodity crops are remunerative and their performance is very much important for farmers. There is need for concerted efforts to enhance production by providing long term incentives to small farmers. The commodity specialized approach assumes that the right way to increase productivity and production of a particular commodity is to group all functions relating to it under one administration. The roles such as extension, research, input supply, output marketing, and extension programme planning are controlled by the commodity organization. Implementation is done through field staff of that organization. Resources tend to be provided by the commodity organization. The success is usually measured on the basis of total production of the particular crop. Extension is fairly centralized and is oriented towards one commodity or crop and the extension agent has multiple functions. There is need to consider many different aspects of agricultural production including food security, poverty alleviation and sustainability.

10.2.3 Project Approach

A project is designed to solve a particular problem of a specific area in predetermined time. The first prerequisite is existence of a problem which has to be solved. This approach concentrates on a particular location, for a specific time period, often with outside resources. Thus, the assumption is that with the help from outside, rural people can substantially improve their lives. In fact, it is like pilot testing of a programme on small scale with all-out efforts for a specific period to see that results are positive and can be up scaled. Part of its purpose is often to demonstrate techniques and methods that could be extended and sustained after the project period. It uses large infusions of outside resources for a few years to demonstrate the potential of new technologies. The case of Intensive Agricultural District Project (IADP), which was known as package Project intended to develop Indian agriculture by diffusing science and technology to the

10.2.4 Training and Visit Approach

Training and Visit Approach was brain-child of Israeli scientist Daniel Benor, which was propagated by the World Bank in many developing nations of the world. It was based on assumption that research-based agricultural messages need to continuously communicated to farmers. Thus, the extension agents need to be updated regularly for effective performance. All this is not possible without a sound management system of training, visit, monitoring, motivation, resource availability, linkages between different stakeholders, etc. This is very intensive approach based on a rigorously planned schedule of visits to farmers and training of extension agents and subject matter specialists. Close links are maintained between research and extension. Extension agents are exclusively involved in technology transfer. Success is related to increases in the production of particular crops or commodities. It has novel approach of choosing contact farmer, who have leadership traits in

fact the system works with limited farmers called as contact farmers. However, this is very costly, centralized and agriculture-centric. The other allied occupations of farmers like livestock, forestry, fisheries, etc. do not get attention of extension service.

10.2.5 Farming Systems Approach

A key characteristic of this type of extension is its systems or holistic approach at the local level. The farming systems approach considers the farm, farm household and off-farm activities in a holistic way to take care of not only farming but all aspects of nutrition, food security, sustainability, risk minimization, income and employment generation, which make up the multiple objectives of farm households. This approach considers interdependencies of the components under the control of members of the household, as well as how these components interact with the physical, biological and socio-economic factors not under the household's control. Close ties with research are required and technology for local needs is developed locally through an iterative process involving local people. Success is measured by the extent to which local people adopt and continue to use technologies developed by the programme. The farming systems approach emphasizes the following:

- that research and extension agendas should be determined by explicitly defined farmers' needs
- an understanding of existing farming is must
- inclusion of farmers and their indigenous knowledge should be ensured
- inclusion of social scientist and women scientists during investigation of problem is must
- researchers, extension workers and farmers work in a team.

10.2.6 Privatised Extension Approach

Now with increasing pressures on the government due to budgetary constraints and international agreements, certain reforms in public sector have become inevitable. There is need for intensive efforts to enhance agricultural production with value addition. Extension services, must come up to the expectations in the changing scenario. Investments in extension are substantive due to requirements of large number of staff, transportation needs and other support services. Under these circumstances, it is being debated whether it would be wise to continue with the present system of extension or choose from available alternatives. While it is essential to reach the masses to create awareness about useful agricultural information, intensive efforts are required for export oriented agricultural enterprises. Privatization of extension does not mean completely closing the government department and handing over extension responsibility to private players. The government will continue to have a greater say in co-coordinating private agencies, monitoring their service and serving as arbitrator in cases of dispute.

There are instances that private firms have played significant role in agricultural extension if policies are favourable. With opportunities in favour, private sector may be willing to provide quality services. In order to serve the resource-rich commercial farmers, fees may be charged for the services whereas the poor may be given free or subsidized services. This may call for careful scrutiny of private firms with competency to provide quality service. Performance standards need to be determined to have check on quality. With advances in information technology, it is possible to use video, video text and computers in providing agricultural information. Satellite mediated communication can facilitate expert advice. Local radio has also emerged as a possibility for communication support. Thus role of public sector extension may include the following :

- To attend to target audiences unserved by private sector
- To coordinate multiple extension providers
- To serve as final reference or arbitrator in case of conflict
- To assure accountability of both public and private extension services to the public.

This approach assumes that technologies have become very complex and extension work by the government is becoming too costly and unsustainable because of large number of farmers to be served. Extension workers are less in number and not very competent. Large and commercial farmers will require more sophisticated technological knowledge and skills not possessed by grassroots extension workers. So to serve them professionally payment will have to be made. Small and marginal farmers may be given subsidized services or free coupons for services. This will change the role of public extension service and more with local people (who do not have the means to pay the full cost) will promote a programme that is more likely to meet local situations and where extension agents are more accountable to local interests. Its purpose is to provide advice and information to facilitate farmers' self-improvement. Experiences in different parts of the world suggest that small and marginal farmers can not pay for extension services. Thus, separate arrangements are required according to financial resources of the clients. In changing situations, there are many organizations available, whose services can be used for providing extension service.

10.2.7 Participatory Extension Approach

It is a new concept that indicates community ownership and involvement in extension process. It refers to the leadership and control of farmers over the process of extension. Farmers have say in design and delivery of extension service. This new approach is farmer oriented and non-prescriptive. It gives equal importance to farmers along with other professionals. While the approach values formal research and development activities, it motivates farmers to try out their own experiments as well. Farmer to farmer extension is also being encouraged through this approach. Problem solving

must begin at the farmers' level. Outsiders have facilitative role. Participatory Agricultural Extension Approach is still in the making. Many efforts are being made to establish such an approach to educate and empower farmers to define and solve their problems and help them take same responsibility. Putting responsibility in the hands of farmers to initiate, implement and evaluate extension programme with support from outside so that Participation means that the people themselves are involved in identifying the problems they face, determining ways to overcome them, designing realistic plans to achieve these goals, and carrying them out. Solutions devised and fulfilled by the people in need are far more likely to prove successful than those imposed from outside. It is believed that people have wisdom to solve their own problems. Since majority of farmers are small and marginal farmers, they can not depend on costly technical solutions. So they have to form groups and conduct extension work suitable to them. Participatory extension provides a framework for extension staff to participate with village communities in facilitating development planning and activity implementation.

This approach ensures that the extension response become community driven and assist village communities implement their planned activities with routine monitoring and evaluation of activities. Success is measured by the numbers of farmers actively participating and the sustainability of local extension organizations.

10.2.8 Educational Institution Approach

This approach uses educational institutions which have technical knowledge and some research ability to provide extension services for rural people. Implementation and planning are often controlled by those who determine school curriculum. The emphasis is often on the transfer of technical knowledge. Implementation and planning are often controlled by those educationist. The emphasis is often on the transfer of technical knowledge. State agricultural universities in India have responsibilities of extension besides teaching and research. There is clear cut goal to take the technologies generated at the University to the people. The area of jurisdiction is limited and selective. Those who are directly involved in teaching and research have opportunity to take the technology to clientele and have first hand experience about the consequences. This is also called as First Line extension system of the country. The scientists are able to improve technologies for better performance based on their experiences.

10.2.9 Market-led Extension Approach

It is an approach through which extension system can provide end-to-end service to the clientele from package of practices for production to selling of produce. Farmers can be facilitated to get remunerative prices for their produces. Farmers have increasingly begun to perceive marketing rather than production as the major constraint to enhancing farm incomes. With major thrust of extension agencies on production techniques, marketing extension

so far has not received the attention it deserves. This assumes greater significance in the light of the new international trading regime under the WTO and opportunities of export available. Public extension functionaries are presently ill equipped to deal with marketing extension. Extension service will need to address these issues through strengthening capacity of the public agency, supporting private sector in marketing extension and making extensive use of media and IT in information and technology dissemination. Marketing extension should be brought centre-stage. Indeed, production will now need to be significantly dictated by market requirements.

Farmers are sensitized on the production aspects like (i) what to produce (ii) when to produce (iii) how much to produce (iv) when and where to sell. Besides this, farmers are also sensitized on consumer preference, market intelligence, processing and value addition and market information system. The market-led extension deals with all these parameters from production to marketing of agricultural produce (Swanson, 2008). Thus, important roles of market led extension are as follows:

- Advice on product planning: Product planning in terms careful selection of crops with market situation in view calls for specific advice to enable farmers to stay competitive.
- Marketing information: Farmers need advice on current price and forecasting of market trends. Information are needed on transportation, location of appropriate market, storage, post-harvest handling and manner of tranjection. Such information should be crop specific, area specific and buyer specific.
- Advice on alternate marketing: In case of glut in the local market, farmers may seek benefit of warehousing. Thus, a planned marketing strategy can help in raising farmers' income.

10.2.10 Cyber Extension Approach

Cyber extension is defined as extension over cyber space, the imaginary space behind the interconnected telecommunication and computer networks (Mishra, 1999). Cyber extension approach includes effective use of Information Communication Technology (ICT), internet, expert systems, information kiosks and computer based training systems to improve information access to the farmers, extension workers, research scientists and extension managers. Internet communications technologies (ICT), particularly mobile phones, are commonly considered a means of improving small holder agricultural development. ICT have aided farmers in:

- Raising awareness of market rates to ensure a fair price for produce;
- Sending orders for goods, especially for livestock and perishable goods;
- Providing up to date weather forecasting;
- Improving access to land records, to support credit applications;

- Increased efficiency / reduced costs of access to extension services.

Check Your Progress 1

Note: i) Write your answers in the space given below.

ii) Match your answers with the answers given at the end of the Unit.

1) Write major features of Generalized Extension Approaches?

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2) What are the urgent reasons for initiating market-linked extension approach?

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3) Why participatory approach is being seen as ultimate solution to most issues of extension?

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10.3 EXTENSION METHODS

There are an array of methods and media available today. An attempt has been made here to discuss commonly available methods and media in terms of their application to extension work. There are three different types of Extension methods. They are as follows:

- Individual Contact Methods
- Group Extension Methods
- Mass Methods

10.3.1 Individual Contact Methods

Farm and House Visit

This is a method of direct face to face contact between farmers and extension workers at farmer's farm or home. This method is specially suited to build acquaintance with farmers and gain understanding of the problems. It is helpful to share information and teaching skills. It is better to maintain regularity in visit if this method is to be used. Farmers should know beforehand about visit. It is better to have system of documentation to note down the transactions made during the visits. Such records serve as easy aid to recall what happened with whom during last visit and plan further.

Office call

It is call made by a farmer or a group of farmers on the extension worker at his office. It is not quite common method in our country. However, it can now be used in many progressive areas. Office call is most appropriate when clients are motivated. It is very economic on time and resources of extension workers.

Telephone call

This has been a popular method in development nations with large network of telephones. This helped farmers to contact the extension agent about their problems and solutions. Telephone callers had to follow pre-decided and other curtsies to get best services.

10.3.2 Group Extension Methods

Demonstration

This is one of the oldest extension methods, equally applicable in many different situations to build confidence of people in new ideas and persuade them to try by showing. Demonstration means to show. Demonstration should not be confused with experiment (trying new ideas under artificial conditions). Demonstration is showing proven techniques to reveal the procedure of doing or value of it. Simply speaking demonstration is a way of showing method (method demonstration) or result of a practice (result demonstration). It is very effective in stimulating learning by involving most senses.

Method Demonstration

Method Demonstration is a method of training in which the trainer shows step by step process of a method/operation and describes it. It is very useful for teaching skills. Skill refers to the ability to do a task. Mastery in skill requires confidence in the technique and efficiency to perform task with investment of minimum time and energy. This can be achieved by showing the correct way of operation in such a manner that one can get confidence to try it out himself. This is where lecture or discussion cannot help.

Result Demonstration

Result demonstration, as the name indicates, refers to showing net worth of a practice or product like new variety, chemical control of pests, etc. It may be for a single recommended practice or a series of practices that come in sequence with respect to a problem. Result demonstrations do not only establish the value of a practice, but also prove the feasibility of the practice under local conditions.

Meeting

The primary purpose of a meeting is to transact business of an organization. Meetings are held for a variety of purposes such as sharing information, solving problems, decision-making or strategy planning. Even though meeting is a common method, farmers do not often open up before outsiders or in groups. Thus, organization of meeting requires relaxed atmosphere to stimulate the silent ones to speak.

Lecturing

Lecture is one of the most common forms of presenting information to an audience, particularly of a large size. Communication is mostly one-way: while the lecture goes on the audience listen and (normally) take notes. It is customary to defer questions (Other than those with a clarifying nature) to the end of the session. Lecture allows a large volume of material to be covered in a relatively short-period of time.

Group Discussion

Discussion refers to verbal interaction between two or more people. Discussion is preferred as a method of teaching because it suits the characteristic of adult learners. Adults have experiences, which need to be shared, explored and analysed to expand their knowledge. Discussion provides them an opportunity to participate in the process of learning. This is an ideal method because the control rests in the hands of learners. Trainers act as coordinators or facilitators. However, the role is very challenging. It requires special skills to create a conducive environment so that learners feel relaxed and motivated to contribute openly.

Field Trip

A field trip is organized excursion for a group of people for the sake of first hand observation of a problem or practice (s) in natural setting. Field trip are organized for a number of objectives as given below:

- It stimulates interest, conviction and action in respect of a specific practice.
- It helps to recognize problems, develop interests and promote adoption.
- It demonstrates feasibility of related practices to the learners in an informal way.

Field Day

Field day is an organized gathering of farmers, extension workers and experts at the site of result demonstration for sharing information and experience about the practice. It is an important extension method to enhance utility of result demonstration. On this day, site of result demonstration is visited by the farmers with the following objectives:

- To learn about feasibility of new practice under local conditions.
- To observe and hear experiences of fellow farmers

Extension Talk

Speech or talk by extension worker to small or large group of audience is quite common method to share information. Formal presentation to provide information or technical skills requires planning.

Checklist for Good Extension Talk

- Speaker has aroused interest of audience in the topic
- He has stated purpose clearly in 1, 2, 3 andterms.
- He moves in organized way
- He explains abstract through examples and arguments
- He uses understandable language
- He uses internal summary every time a major point is made.
- He gives overview, states the point supporting it and reviews them in the end.
- He signals transition from one point to another.
- He uses gestures.
- He adapts pace of delivery
- He changes volume of speech rhythmically
- He exuberates confidence and enthusiasm
- He uses summary to wind up the talk.

Buzz Group

This technique is a modified form of brainstorming, which aims to generate discussion and to lead to new ideas and solutions on the basis of small group discussions. A buzz group normally consists of about 5-6 members. The group may be asked to produce ideas on a narrow or an open-ended topic within a specified time limit. It is customary for the group to appoint a reporter to convey their findings at the end of their deliberations.

Brainstorming

Brainstorming is a method used in small groups to generate creative ideas without inhibition. It is helpful to break the monetary and use resources/

experiences of participants in solving a problem. Unlike group discussion and problem solving, emphasis is on encouraging every member to come out with ideas without criticism at the beginning. The idea is to benefit from diversity of experience, collective thinking and synthesis of contribution in an open and accepting atmosphere.

Exhibition

Exhibition is systematic display of information with the help of variety of methods and media of communication in order to create interests and desire among visitors to adopt. Exhibition is arranged for a variety of purposes as given below.

- Exhibition is arranged to create public interest in new technologies.
- It helps to create goodwill and gain publicity.
- It draws large crowd if coincided with local festival and/or fairs.
- Use of local material and involvement of farmers will create enthusiasm.
- It influences people to new practices.

Extension Campaign

Campaign is concentrated use of different communication methods and media around a theme for a period of time. It is well organized and intensive activity. A campaign creates wide public awareness and provides specific message. Dramatic use of different channels ensures attention and interest.

Mass Media

Radio

Radio is most popular mass media, among majority of rural people. Transistor radio sets are cheap and convenient to use. There is no problem of electricity maintenance and repair.

Special Features of Radio

- It is cheap.
- It is portable.
- It provides information entertainment and education.
- It can quickly transmit messages in most remote areas.
- Distance learning is possible through radio.
- It is helpful in forming public opinion.
- Listener can get the message even while engaged in other activities.

Local Radio/Community Radio

Looking to the popularity of radio as medium of communication and its effectiveness in development, All India Radio (AIR) ventured into a new phase of broadcasting by experimenting with the concept of local radio

station. A local radio station serves a small area (a district or so) with similar agro-climatic and cultural situations. The programmes are supposed to reflect local culture and aspirations. They are supposed to support on-going developmental programmes. Field-based programmes using local talents, give voice to people's views. Local culture finds more air time. Community service programmes should provide opportunity to people, their organization and development organization to broadcast matters or information. Radio thus becomes voice of the people and catalyst in development. Unlike regular radio, extension workers can find air time.

Audio-Cassette Technology (ACT)

Tape recorder or audio-cassette tape is now very medium in Indian homes for listening to music. The technology has become cheaper and accessible. Experiences in many developing nations suggest new possibilities for use in extension work. ACT gives audience the choice regarding content of message. Technical messages can be designed systematically keeping the characteristics of audience in mind. Pre-recorded talks of technical experts may compensate for lack of expertise of grassroots extension workers.

Television

Television has unique advantages over other mass media. While it provides sound, vision and movement. It can reach largest number of people in shortest possible time. It is a medium unlike any other. TV screen is small and depends on close-ups. People can watch sitting in their homes. Television viewing does not demand extra strains to go out or read a book and the message are pre-selected, sorted out to present in simple manner. The medium is quite suitable for subjects that require dramatized presentation, identification of objects, live demonstration of complex technical process depicting animated presentation and presentation of experiences, places, processes unfamiliar to viewers.

Video

Video is an important aspect of advancement in communication technology. It provides the facility of audio-visual communication like film or television but with less complexity and added advantages. Video is unique medium for audio-visual communication. It is decentralized and thus useful to communicate local messages among decentralized and thus useful to communicate local messages among groups with similar characteristics. The images are sharp unlike films. Instant playback facilities for pause and replay at any time allow viewers to review materials at will.

Circular letter

This is a letter reproduced with same content and information to many people periodically or on some occasions. It is good way to keep regular touch with farmers and to communicate timely information about in a personal way. It is a good way to provide information about new technologies to farmers

quickly. It is also very helpful to invite farmers on the occasion of field day, demonstration, exhibitions, etc.

Rural Publications

Printed words carry credibility. Increasing rural literacy has widened the scope of using printed materials for the farmers. Even though majority of farmers is illiterate, there is always someone in the immediate environment to read out to them. Thus, potential consumers of extension publications are more than number of literates. There are other advantages of using publication, as below:

Farm Journals

Farm journals are considered as important tools to disseminate relevant development messages to the rural audience.

Writing for Rural Readers — Some Tips

Most reading materials meant for normal readers with education and exposure to printed media, do not fare well with readers. Writing for people deeply rooted in oral culture with very little or no exposure to written words is a challenge. It calls for innovations in written communication. Some of the key-points needing attention is given below:

- The theme/topic of writing should be intimately related with the life and problems of the readers.
- The idea to be communicated should be kept as simple as possible. The title should be catchy.
- The idea should be introduced in part by informing about the contents in advance.
- Each major point should be covered separately giving suitable subtitles.
- It is better to support text by giving matching pictures or line drawings, side by side.
- Use examples and analogies easily understood by the target group.
- Repeat the key points in the end and even underline them.

Indigenous Communication Channels

In every society there are various forms of communication among people. Some channels and forms of communication are deeply rooted in the culture and preserved traditionally from generation to generation. Such channels are called as indigenous traditional folk media. They serve various social needs of the community. They are quite, face to face and linked with emotions and values of people. Thus, they are quite powerful in raising consciousness of people. They are cheap and do not require external resources. Examples of Indigenous communication channel may include various social gatherings like feasts, village meetings, spontaneous gathering at tea shops, oral

narrations, etc. Mundy and Compton (1991) have stated that indigenous communication can take many forms including the six given below:

Use of Indigenous Communication in Extension

Indigenous channels have attracted attention of social scientists as diffusion studies indicated importance of opinion leadership and inter-personal networks. Of late, studies have indicated how new varieties of agriculture were adopted by the farmers without help or deliberate efforts of extension workers. Peer group, village elders, youth groups, market place and ceremonial strategically. Experienced farmers can be used as trainees.

New Communication Technologies

New communication technologies have emerged such as microcomputers; cable TV, Teleconferencing, teletext and desktop publishing are all examples of advances in communication technology. Many of these technologies use miniature forms of erstwhile larger equipments and offer opportunities of interaction viz. Computer-assisted-instruction, interactive video, and compact-disc read-only memory (CD-ROM).

Micro-computers

Computers are classified as mainframes, minicomputer and microcomputer based on their power, speed of executing a set of instructions and the amount of memory available to store data as well as the degree of inter connectivity the computers offer. Mainframe computer were large and costly. Mini computers are less powerful. Microcomputers are based on micro-chips technology.

Electronic Mail (E-Mail)

Electronic mail or E-mail is a system which passes information electronically from sender to receiver. Thus, it is a simple way to communicate between two people through computer. E-mail is like surface mail in which a message or letter is typed into computer and sent to another computer via telephone. The message is stored there until the receiver asks it. Thus, computer acts here as post office. The message in E-mail can be in one of the many forms voice to text image or drawing. An E-mail system provides for mail box which user can see by using personal computer and modem connected to telephone. E-mail has following advantages.

Interactive Video

Interactive video refers to a video system which is a combination of computer and video. It uses multi-media approach i.e. text, stills, video, audio, slides, overheads, etc. The messages stored in various forms are retrieved by user according to choice.

Teleconferencing

Teleconferencing is interactive group communication. Teleconferencing refers to a system for creating dialogue between geographically dispersed participants. Advances in telecommunication and computer have made it

possible to hold meeting without traveling to long distances. Teleconferencing can help physically distant people talk together.

Types of Teleconferencing

Audio teleconference

The simplest form of teleconferencing is the use of amplified speaker phones to join groups of people at each end of a telephone conference. High quality audio reception and transmission can be achieved.

Computer Conferencing

Computer conferencing is a variation of electronic mail which enables multiple users, geographically isolated to carry out dialogue. Information is transmitted through a computer with an elaborate store-and-forward message handling system. In a computer conference participants need not be involved simultaneously. One participant may send a message at any time and it may be read by others whenever they have time. A computer conference is basically a group activity. A group can select to participate. Each participant may put their message and read others' contribution. Messages are stored, thus allowing those who join late to retrieve and catch up with the conference. Video conferencing is like television.

Multi-media Strategies

Studies have proved that there is no one best medium of communication. No single medium can effectively meet the goals because each medium has its own characteristic strengths and weaknesses. However, combination of media can help in complementing and supplementing each other. The strength of certain media can help to compensate the weaknesses of other media in combination

Check Your Progress 2

Note: i) Write your answers in the space given below

ii) Match your answers with the answers given at end of the Unit.

1) Is Office call method suitable for extension work in India?

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2) Why should we choose mass media for agricultural extension?

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10.4 LET US SUM UP

Extension approaches have emerged in response to the diverse socio-political situations in different countries. Generalized Agricultural Extension system is most popular across the nations. Training and visit extension evolved out of weaknesses of traditional extension systems. However, due to elaborate infrastructure, high cost, it could not be sustained. There is ray of hope for sustainability of extension system in participatory Agricultural Extension Approach. A number of extension methods are being used. Group methods are most suitable and popular in general. However, mass media are gaining prominence. New communication technologies have helped in reaching farmers useful messages at the doorsteps.

10.5 KEYWORDS

Project: It is a short duration activity with specific objectives and concentrated efforts to achieve them within time.

Participation: It refers to intensive involvement of people at finding and analysing problems and finding opportunities to work together to achieve goals.

Privatization: It refers to taking over the works of the government by private business houses/people.

10.6 SUGGESTED READINGS/REFERENCES

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10.7 CHECK YOUR PROGRESS- POSSIBLE ANSWERS

Check Your Progress 1

- 1) The major features of Generalized Extension Approaches are as follows:
 - This is an approach to disseminate available technologies to the farmers.
 - This is also known as top-down approach.
 - A large structure of extension agents needed to run extension services.

- 2) Market -Led-Extension approach has emerged in response to the market forces after economic liberation. Commercialization and free trade economy demand qualitative and quantitative increase in production of food products. Hence farmers need to know about standard practices, market, interaction trade etc. Thus extension has to be linked with market.
- 3) Because participatory extension approach is for the farmers, by the farmers and of the farmers. It takes into consideration farmers' problems, resources and capacities. Peoples, organizations have to be created and sustained. This will overcome problems of farmers' dependence on the government extension agency.

Check Your Progress 2

- 1) With proliferation of commercial agriculture in India there are number of progressive farmers interested in using latest technologies and grow crops of standard quality. Export oriented agriculture also calls for new knowledge and technologies. Thus, farmers need latest information and proven technologies and practical training. Thus office calls becomes very significant. Agricultural clinics being established by government of India are also model on office call pattern.
- 2) We choose mass media for agricultural messages because of the following strengths of the media:
 - Speed: They are fast.
 - Coverage: They cover a large area.
 - Audio visual appeal: They have both audio and video appeal to the masses.
 - Multiple effects: People spread messages by word of mouth.
 - Economic: It is less costly because of large area coverage.

UNIT 11 DIFFUSION AND ADOPTION OF NEW TECHNOLOGIES

Contents

- 11.0 Aims and Objectives
- 11.1 Introduction to the Concept of Diffusion of Innovation
- 11.2 Innovation
- 11.3 Adoption of New Technologies in Development Process
- 11.4 History of Diffusion of Innovation
- 11.5 New Technologies
- 11.6 Adoption of New Technologies Innovation
- 11.7 Adoption Process
 - 11.7.1 Adoption Categories
- 11.8 Adopter Categories on the Basis of Innovative
- 11.9 Importance of Opinion Leaders
- 11.10 Characteristics of Innovation
- 11.11 Let Us Sum Up
- 11.12 Key Words
- 11.13 Suggested Readings
- 11.15 Answer to Check Your Progress

11.0 AIMS AND OBJECTIVES

This unit deals with the importance of diffusion and adoption of new technologies in the development process. We shall talk about the history of diffusion of innovation and the concepts of new technologies. This module provides a systematic explanation about the diffusion and adoption of new communication innovations.

At the end of this unit, you will be able to:

- Describe history of diffusion of innovation;
- Explain the meanings and components of new technologies;
- Discuss the process of diffusion of new technologies;
- Analyse the significance of diffusion of new technologies in overall growth;
- Describe adoption process;
- Discuss categories of adopters; and
- Explain characteristics of innovations.

11.1 INTRODUCTION TO THE CONCEPT OF DIFFUSION OF INNOVATION

The concept of diffusion of innovation with references to technological advancement and ideas was identified by Everett Rogers (1962). He argued about significant role of communication in disseminating information for development. Interestingly, with passing of time the perspectives on diffusion of innovations theory have changed as initial concept was unable to explain development appropriately.

- As explained by Rogers (1962), 'diffusion' is the process through which an innovation is communicated via definite channels over time among the audiences of a social system.
- It is a special type of communication wherein the messages are specifically concerned with new ideas. We are aware that communication is a process for sharing information with one another for reaching a level of mutual understanding. Thus, communication is basically a process of convergence as two or more individuals are involved in exchanging information.
- Diffusion is a social process that occurs when people learn about an innovation. In its most basic form, diffusion involves an innovation that is communicated over time among members of a social system via specific channels. The time of adoption is a common dependent variable in diffusion. On the other hand, adoption occurs when a person does something different than they did previously, such as purchasing or using a new product or technology, learning and performing a new behaviour, etc. The key to adoption is for the individual is to perceive the idea, behaviour, or product as novel or innovative. Diffusion is possible as a result of adoption to a great extent. Diffusion has very appropriately been defined as a kind of social change and the process through which adaptation follows in the structure and function of the social system. It is believed that when new ideas are invented and diffused, and is either adopted or rejected, they eventually lead towards certain consequences and at that point of time social change occurs. Well, there are possibilities that such change can take place in other ways too, for instance, through a socio-political revolution or through natural calamities. The word 'diffusion' includes adoption of both the strategic and the natural spread of new ideas.

11.2 INNOVATION

An innovation is basically as an idea, or object which is perceived as new by an individual, organization or any other unit of adoption. Interestingly, the perceived newness associated with the idea of the individual decides his or her reaction to it. If the idea appears new to the individual, it is considered

innovation. 'Newness' in an innovation need not necessarily just include new knowledge. The newness of an innovation could be conveyed in terms of knowledge, influence, or a decision to adopt. Well, it should not be presumed that the diffusion of all innovations including technologies are essentially desirable. In fact, there are some studies indicating towards harmful and uneconomic innovations which are generally not desired by the larger social system. An innovation always has two components – the hardware and the software. This is clear in the case of the computer where the machine (hardware) is useless without the programmes which instruct it what to do (software). It is also true for a plant variety where we have the plants (equivalent to hardware) and the techniques for growing them (equivalent to software). It is possible that the same software can be used for a new variety as for the old variety, but we often need new techniques also for cultivation, fertilizer application, etc to ensure optimal production by the new variety. Farmers often play an important role in developing the right kind of software. Scientists who develop new hardware in their research station should take into account the software which might be available to farmers. However, they often fail to do this. For example, they have developed many more techniques for irrigated agriculture although no irrigation is available for the last majority of farmers.

11.3 ADOPTION

It can be defined as the process through which a person can reach at the decision whether to accept or reject an innovation or idea after becoming aware of it. Rogers (2003) stressed on the role of mass media in creating awareness among the people and distinguishes that apart from the mass media, it is the personal interest and sources that eventually influence the adoption process of changes. The early diffusion research studies were precisely directed towards the role of mass media in creating awareness. Diffusion is a type of social change defined as the process by which a social system's structure and purpose change. New ideas are created, disseminated, and adopted or rejected, resulting in specific outcomes in social change. Of course, such change can also happen through other means, such as a political revolution or a natural disaster like a drought or an earthquake.

11.4 HISTORY OF DIFFUSION OF INNOVATION

Since the 1940s and 1950s, study on the diffusion of innovations started taking place in sequences. It emphasized on the ability of media messages in creating knowledge about new ideas and practices. The theory of diffusions of innovations evolved as an indigenous agenda for modernization development and stressed on the role played by opinion leaders in persuading the target group in adopting the technological innovations in a society. This theory also emphasized on the role of communication process and communication effects. Researchers have studied the diffusion of innovation

from diverse perspectives. For example, educational scholars have studied innovation and adoption in the context of new pedagogical ideas while the rural sociologists have considered innovations in the context of adapting to agricultural advances in farming process. It is quite clear from the above statements that diffusion of innovation research had been studied in different contexts and disciplines based on their applicability and outputs. By looking at the work ‘The Passing of Traditional Society’ by Daniel Lerner (1958), we would be able to understand the communication perspective based on references and theories that are collectively assembled in diffusion paradigm. Daniel Lerner’s work “The Passing of Traditional Society - Modernizing the Middle East” is noted to be defining diffusion paradigm and at same time marking the introduction of development communication. This notable work by Lerner (1958) is deliberated as the architect of change from a traditional conservative society to a modern technological society possible via mass media.

The role of communication in the development process was acknowledged by experts and social scientists long back. In his famous treatise ‘The Passing of the Traditional Society’ by Lerner (1958) recognized the role of communication in the development process and that the evolution of mass media had been one of the three phases of democratic and political progress. Mass media has a significant role to play in the process of development. We are already aware of the way with which mass media had led directly to socio-economic and cultural improvements across the developing nations of the world. In this context, Schramm (1964) made an observation stating that mass media offers the vast scopes of new approaches of transforming and uplifting the lives of the deprived people. In other words, this phenomenon of information dissemination is termed as development communication by scholars.

Till the early 20th century, researchers studying diffusion of innovations theory differed on the question of whether an innovative idea developed autonomously according to different cultures or whether they were invented in one culture and diffused or adopted by another culture.

11.5 NEW TECHNOLOGIES

A technology can be defined as a design for contributory action that decreases the ambiguity in the cause-effect processes involved in attaining a desired result. In simpler words technology usually has two components – firstly hardware aspect that comprises of the tool that represents the technology as physical objects, and secondly the software aspect that consists of the information base for the physical tool. For instance, ‘computer hardware’ comprising of transistors, electrical connections, semiconductors, and the metal frame for protecting all these electronic components, and on the other hand we have ‘computer software’ comprising of the instructions and coded commands that allow us to use it to encompass human competences in

solving various problems. These two aspects of technological innovations make us understand the close interaction between a tool and the way it is being out to use. Even though the software constituent of a technology is often not so obvious to observation, it should be noted that ‘technology’ in its complete sense always signifies a mixture of both hardware and software aspects.

Technologies encompass conventional technologies to the most modern form of tools. The tools range from radio, television to telephones, mobile phones, computers and also Internet. Mostly modern technologies include all software systems, electronic networks, communication devices as well as applications that lead to increased information transmission within less time and low costs. It is made possible by modern technologies that we are able to communicate in real-time with others across the world using instant messaging, voice message, video-conferencing and much more. We all are experiencing it in our daily lives on a regular basis. It is undeniable that rapid innovations in the technological front are making communication technologies less expensive and easier to access, so in a way bringing its full potential within the reach of a bigger number of people. So, in summation it can be stated that most technological innovations are produced by scientific research activities, although they often are a result from are relationship of scientific method and practical setups.

Check Your Progress 1

Note: i) Use the space below for you answers.

ii) Compare your answers with those given at the end of this unit.

1) What is meant by innovation?

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2) Write in brief the significance of diffusion?

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11.6 ADOPTION OF NEW TECHNOLOGIES IN DEVELOPMENT PROCESS

There are a number of areas where technology adoption has led to tremendous development such as agriculture, medicine industry, finance, infrastructure, governance, education, etc. to count a few. There is a connection between technology and innovation. In simpler words, it can be said that the adoption of technology can result in innovation that eventually justifies the goal or objective of improvements by any institution, individual or the larger society. The contribution of new technology in overall growth can be realised only when and if it is widely disseminated and used. Diffusion occurs as a result of a series of individual decisions to begin using the new technology, decisions that are frequently the result of weighing the uncertain benefits of the new invention against the uncertain costs of adopting it. We can appropriately say that Rogers' 'diffusion of innovations theory' is the most suitable for examining the adoption of technology in various developmental sectors like higher education, agriculture, medical and more. In fact, much of the diffusion of innovation research involves technological innovations and hence both the words "technology" and "innovation" are largely used as synonyms. Generally, innovations that are perceived by audiences as having larger relative advantage, less complexity and compatibility would be adopted much rapidly than any other innovation. This statement lies appropriate for the technological adoptions.

11.7 ADOPTION PROCESSES

Research studies have clearly demonstrated the extensive delays that often occur between the time farmers first hear about favorable innovations and the time they adopt them. It takes four years on average for the majority of mid – Western US farmers to adopt recommended practices. Research workers have naturally have been to find out what happens during this time. The following stages are often used to analyze the diffusion process:

- 1) Awareness – One first hear about the innovation
- 2) Interest - Seek further information about it.
- 3) Evaluation – Weigh up the advantages and is advantages of using it.
- 4) Trial – Test the innovation on a small scale for yourself.
- 5) Adoption – Apply the innovation on a large scale in preference to old methods.

The adoption process does not always follow this requence in practice. For example, it is not feasible to test a new farm building on a small scale. Interest may precede awareness.

An identical stage to the above could be

- 1) Knowledge
- 2) Persuasion (forming & changing attitudes)
- 3) Decision (adoption or rejection)
- 4) Implementation
- 5) Confirmation

The mass media like newspapers, magazines, radio and television generally can play a role in the above processes.

11.7.1 Adopter Categories

It is understandable that not everyone adopts innovations at the same rate. Some people accept new ideas years before others. People are often divided into five categories according to their scores or an adoption index. These are: (in per cent)

- | | |
|-------------------|------|
| 1) Innovators | 2.5 |
| 2) Early adopters | 13.5 |
| 3) Early majority | 34.0 |
| 4) Late majority | 34.0 |
| 5) Laggards | 16.0 |

These percentages serve mainly to make findings from different studies comparable by using a uniform classification. It is more important for extension agents that a minority of farmers are either early adopters or laggards, and a minority of the early adopters are innovators.

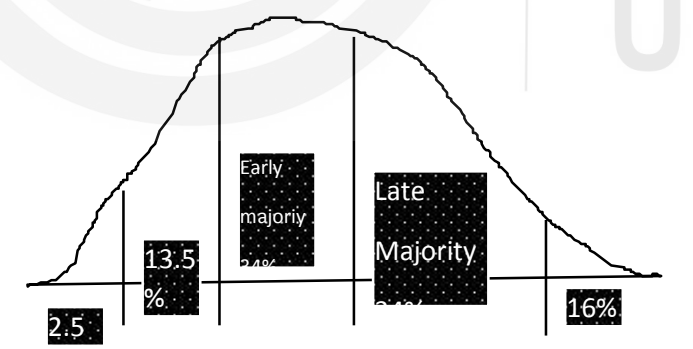


Fig. 6.1: The diffusion curve

11.8 ADOPTER CATEGORIZATION ON THE BASIS OF INNOVATIVENESS

- 1) Innovators – They are the first few to adopt a new idea or practice. They like to try out new ideas or practices. They are daring to take risk. They are also ambitious to achieve success. Normally, they are educated and have contact with sources of information out side the social systematic

make use of cosmopolite sources of information. They have control of substantial financial resources. They also have ability to understand and apply complex technical knowledge. However, they are deviant because they differ in many ways from the average farmers. They can not be role model for the average farmers.

- 2) Early adopters – (Respectable – respected by this peers) – They come next to innovators to adopt new idea or practice. Potential adopters look to early adopters for advice and information about the innovation. The early adopter is considered by many as “the man to check with” before using a new idea. Because early adopters are not too far, a head of the average individual in innovativeness, they serve as a role model for many other members of a social system.
- 3) Early majority – (Deliberate – considering carefully). They are the ones who adopt a new idea or practice a little earlier than the average person. The early majority may deliberate for sometime before completely adopting a new idea. “Be not the last to lay the old aside, nor the first by which the new is tried”, might be the motto of the early majority.
- 4) Late majority – (skeptical – doubting attitude). They are the ones who adopt innovations first after the average person. There are cautious and do not adopt until most others in their social system have done so. The pressure of peers is necessary to motivate adoption.
- 5) Laggards – (Traditional and conservative) laggards are the last to adopt an innovation. They are the most localite in their outlook and many are near isolates. Their point of reference is the past. Laggards tend to be frankly suspicious of innovations, innovators, and change agents.

11.9 IMPORTANCE OF OPINION LEADERS

Opinion leaders have considerable influence on the way in which people in their village think and farm. Farmers interact and learn more from friends, neighbours and relatives. They observe and learn from successful results of known farmers. They keep track of successful results of their neighbours and relatives and seek their advice. Some of these successful or progressive farmers are willing to share their experiences with other farmers. In this way they become opinion leaders in the village because they help other farmers solve problems. Farmers in an increasing number of villages like to know which new farm practices they should adopt and which they should not. People who are well informed about these practices tend to become opinion leaders. Different types of opinion leaders will emerge in these villages. An opinion leader will fulfil several of the following functions in his group with regard to innovations

- Pass on information from outside the group
- Interpret outside information on the basis of his own opinions and experience;

Sets an example for others to follow.

- “Legitimizes” or rejects changes that others want to carry out. That is to say, he gives his approval or disapproval for these changes; and
- Is influential in changing group norms. Not all opinion leaders will do all of these things.

Opinion leaders are able to solve many problems faced by their people. Hence, other villagers may follow their advice or example in the belief they will be successful as well. These leaders may exert influence on a wide range of topics in traditional societies. They may also have considerable power over collective decisions.

11.10 CHARACTERISTICS OF AN INNOVATION

- 1) Relative advantage – Does the innovation enable the farmer to achieve his goals better or at lower cost than he could previously ? This advantage can be influenced by giving incentives to the farmer, such as by providing seeds at subsidized rates. Such incentives may motivate farmers to try the innovation.
- 2) Compatibility with socio-cultural values and beliefs, with previously introduced ideas or with farmers felt needs. Clearly it is very difficult to introduce Pig husbandry among Muslims even if it is a very profitable enterprise. On the other hand, farmers who have received large yield increases by growing improved wheat varieties are likely to be very happy about accepting improved rice varieties as well. However, if an innovation fails after introduction it will be very difficult to get similar innovations adopted.
- 3) Complexity – Innovations often fail because they are not implemented correctly. Some require complex knowledge or skills. For example, it may be necessary to introduce a package of several relatively simple but related innovations. Each on its own is easy, but the relationship between them may be difficult to understand. Dairy cows with a higher genetic potential will produce more milk only if they have feed which is higher in protein and energy content. This in turn will require different crop husbandry practices.
- 4) Trialability – A farmer will be more inclined to adopt an innovation which he has tried first on a small scale on his own farm, and which proved to work better than an innovation he had to adopt immediately on a large scale. The latter involves too much risk. Trialability may be related to divisibility, as with fertilizers, for example. Although large machines cannot be divided; sometimes they can be rented before they are purchased.
5. Observability – A farmer can see a mile away that a colleague has changed from fodder beets to maize as cattle feed, but he might not know the book – keeping system used by his next door neighbour. Out of fear

for the “evil eye” a farmer might not show his improved cattle to his neighbours. Farmers learn much from observing and discussing their friends experiences. Their observations are often a reason to start these discussions.

Check Your Progress 2

Note: i) Use the space below for you answers.

ii) Compare your answers with those given at the end of this unit.

1) Discuss briefly the role of ICT in development.

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2) Do you think that technological innovations have uncertainty associated with it?

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11.11 LET US SUM UP

So, in summation we can say that diffusion is considered as a distinct type of communication, where the contents are concerned with a new idea or innovation. It is specifically this innovation of the idea in the message of communication which gives ‘diffusion’ its distinctive character. The newness also suggests that some level of uncertainty is involved in the diffusion of innovations. The concept of diffusion of innovation has been studied from different perspectives. Furthermore, the idea of adoption could also be defined from a different perspective as the after becoming aware of an invention or concept, the procedure through which a person might decide whether to embrace or reject it. The chief fundamentals in the diffusion of new ideas included an innovation or technology which is communicated through various channels among the members of a social structure. An innovation is defined an idea, practice, or object conceived as new and appropriate for adoption by an individual or other unit of the society. Almost all the ideas discussed in this module are about technological innovations. A technology is simply defined as an outline for instrumental action that decreases the uncertainty in the process involved in attaining a desired outcome. Most technologies have generally two components – hardware and software. This module also gave us an understanding about how technological diffusion in the existing processes of upliftment and

development in marginalised societies. We have also observed how any technological innovation also generates some level of ambiguity because of its originality to the individual. This is largely known as evaluation of diffusion of innovation and its subsequent consequences. This module also made it clear how diffusion of innovations research has occurred in recent years as a sole yet unified body of conceptions and overviews in social science, even though the studies are continued to be undertaken by researchers in numerous scientific disciplines as well.

11.11 KEYWORDS

Diffusion: Diffusion is the process by which an innovation is communicated to members of a social system over time through specific channels. It is a unique form of communication in that the messages are about new ideas.

Innovation: An innovation is an idea, practice, or object that is perceived as new by an individual or other unit of adoption.

ICT: Information and Communication Technologies (ICTs) is a broad term for Information Technology (IT), which encompasses all communication technologies such as the internet, wireless networks, cell phones, computers, software, middleware, video conferencing, social networking, and other media applications and services that allow users to access, retrieve, store, transmit, and manipulate data in a digital format.

Technology adoption: The successful integration of new technology into any firm or development process is known as technology adoption. Adoption entails more than simply putting technology to use.

11.12 SUGGESTED READINGS

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11.13 ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress 1

- 1) An idea, practise, or object that is perceived as novel by an individual or other unit of adoption is referred to as an innovation. In terms of human behaviour, whether or not an idea is 'objectively' new as measured by the time since its first use or discovery makes little difference. The individual's reaction to the idea is determined by how new it appears to him or her. It is an innovation if the concept appears novel to the individual.
- 2) Diffusion is a type of communication in which the messages are about a new idea. It is the novelty of the idea in the message content of communication that distinguishes diffusion. Because it is new, there is some degree of uncertainty involved.

Check Your Progress 2

- 1) The ICT revolution has opened up a plethora of new economic and social revolutions that can assist any country's development process. Its appeal has grown as a result of the convenience and reasonableness with which it may use ICT technologies. ICT is far more than social networking sites, and it can be wisely applied to support numerous elements of development such as education, business, health, government, and so on.
- 2) In the minds of potential users, a technological advance causes one sort of uncertainty about its projected implications, while also providing a possibility for decreased ambiguity in another sense, that of the technology's knowledge base.

UNIT 12 RURAL EXTENSION: INNOVATIONS AND EXPERIENCE

Structure

- 12.0 Aims and Objectives
- 12.1 Introduction
- 12.2 Concept of Rural Extension
- 12.3 Principles of Rural Extension
- 12.4 Extension Approaches
- 12.5 Innovations, Reforms and Experiences in India
 - 12.5.1 Support to State Extension Programmes for Extension Reforms
 - 12.5.2 Extension Through Educational Institutions
 - 12.5.3 New Initiatives of ICAR
 - 12.5.4 Extension Through NGO
- 12.6 Let Us Sum Up
- 12.7 Suggested Readings
- 12.8 References

12.0 AIMS AND OBJECTIVES

The unit aims at introducing the concepts, principles of extension, rural development, and role of extension in rural development. We shall make you familiar with the definitions of extension given by various authors, academicians and experts along with their views on extension. This unit will also aim at clarifying the general and specific objectives of extension.

After studying this unit, you should be able to:

- Write various definitions of extension given by different authors, academicians and experts along with their views on its concept;
- State the concepts of rural extension and rural development and principle of extension;
- Discuss the reforms and innovations in extension; and
- Try to relate extension and rural development and to know how extension helps in rural development.

12.1 INTRODUCTION

Rural development has been an avowed task of the government ever since the beginning of planned era. Its policies intended to improve the quality of life and well-being of rural people for which agricultural development is highly imperative. India is predominantly an agrarian country and a vast majority of

people are rural dwellers and are dependent on agriculture for their livelihoods. Agriculture is a highly significant sector of Indian economy with contribution of 14.4 per cent to GDP (FY19-20) and engaging about 41.49 per cent of workforce. Gandhi jee (1936) had rightly stated that if village perished, India would perish too. Not only in India, world-wide, has the welfare of the rural population been linked to agricultural development. An analysis of rural development programmes in various democratic countries reveals that extension education process has been the most effective means for ushering development. The essence of the process has been helping people by means of education to put useful knowledge to work for them. It aims at teaching them 'how to help themselves' with a minimum of government aid. It lays emphasis upon working with them rather working for them. It strives to change the behaviour of people in such ways that they 'help themselves' for their economic and social improvements.

It has been experienced that the key to rural development in a free society is the human element, not material aid. What leads to an enduring change is the education of the people to do things for themselves. The working procedure of rural extension may be defined as "helping the rural population to help themselves" (Panders, 1959). The determinant of success is not merely planning and launching a programme to promote change among rural people, but how the people make response to it. People's participation in the programme is crucial. The central means of rural progress is a people programme with government aid.

There is growing emphasis upon emancipation of people from ignorance, hunger, poverty and disease through self-help. People have innate traits and capacities that potentially can enable them to lift themselves to a higher order of economic and social well-being. They just need to be motivated, trained and supported with such critical inputs and necessities for achievement which they cannot afford themselves. Extension aims at teaching the people about new methods of doing things and help them to put in to action the learned technologies for shifting away from business as usual approach. The experiences underline that it is the human is the central element not the technology, which propels or retards progress in rural India.

It is pertinent to linkup how extension helps in rural development. Attainment of Sustainable Development Goals (SDGs) set by United Nations is one of the prime tasks of all nations and extension could play a major role in realization of the major goals. There are 17 Sustainable Development Goals viz, No poverty; zero hunger; good health and well-being; clean water and sanitation; affordable and clean energy; decent work and economic growth; industry, innovation and infrastructure; reduced inequality; responsible consumption and production; climate action; life below water; life on land; peace and justice strong institutions; partnerships to achieve goals. Agricultural development is directly related to a majority of the above goals.

Rural Extension interventions could foster and strengthen capacities of rural communities for collectivization and building social capital; while awareness campaign, training and capacity-building of rural communities would lead to motivate and capacitate to eschew practices which lead to greater GHG emissions and to adopt climate smart technologies for adaptation and mitigation. Rural women make significant contribution in the agricultural activities. Gender mainstreaming is one the salient tasks of extension and it would reduce inequality in society. The principle of leadership of extension would promote rural leadership and participation in planning and execution of rural development programmes. Thus it can be understood that rural extension could be an effective means for rural development.

12.2 THE CONCEPT OF RURAL EXTENSION

According to FAO, extension is concerned with agricultural education aims at assisting rural people to bring about continuous improvements in their physical, economic and social well-being through individual and cooperative efforts (Kamath, 1961). The objectives of rural extension are to raise agricultural productivity and rural welfare (Panders, 1959). The fundamentally extension aims at developing the rural people economically, socially and culturally by means of education.

It is believed that the use of the term 'extension' originated in England in 1866 with a system of university extension which was taken up first by Cambridge and Oxford Universities, and later by other educational institutions in England and in other countries. The word Extension is derived from the Latin roots: *ex* meaning out; and, *tensio* meaning stretching.

- 1) According to Rolling (1986), Extension refers to the actual work done by professional extension agents at various levels of management; while Extension Education refers to the body of knowledge concerning that work or practice.
- 2) Extension involves the conscious use of communication of information to help people form sound opinions and make good decisions (Van den Ban & Hawkins, 1998).
- 3) Extension is everything that people who think of themselves as extensionists do as part of their professional practice (Leeuwis, 2004).
- 4) Extension is also defined as the increased dissemination of useful knowledge for improving rural living.
- 5) In its broad sense, extension refers to an informal educational function that relates to any organization disseminating information and advice to promote learning, although it tends to be associated with agriculture, fisheries and aquaculture, and rural development in general (Rivera & Qamar, 2003)

Extension has been perceived primarily as a means of extending or disseminating information and ideas to rural people in order to bring desirable changes in human behaviour and development can be seen through the changes brought in. Extension is the catalyst for affecting development by bringing changes in knowledge (things known), skills (things done) and the attitude (things felt).

Extension is a concept that is subject to a wide range of interpretations. The term extension has been interpreted in the following ways:

- Extension is an informal method of education targeted at the rural population. This method provides suggestions and resources to help them overcome their problems. Extension also intends to boost the productivity of the family farm, enhance production and, in particular, increase the quality of living of the rural people.
- The main objective of the extension is to change the attitude of farmers towards their difficulties. Extension is concerned not only with physical and economic accomplishments, but also with the growth of rural people themselves. Extension agents therefore address issues with rural people, help them gain a better insight into their difficulties and also help them determine how to solve them.
- Extension is a process of collaborating with rural people to enhance their livelihoods. This focuses on helping farmers to enhance the productivity of their agriculture and also improving their capacity to direct their own future growth.

They all underline extension as an educational process which works with rural people, supports them and prepares them to confront their problems more successfully. Ensminger had rightly said that extension is an education and its purpose is to change the attitude and practices of the people with whom the work is done. The concept of extension has combined the concern of philosopher and philanthropist for development of whole man with the methodology and programmes of change.

Check Your Progress 1

Note: i) Use the space below for you answers.

ii) Compare your answers with those given at the end of this unit.

1) What do you understand by the term EXTENSION?

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2) What is the significance of Extension in rural development?

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12.3 PRINCIPLES OF EXTENSION

We can list various principle of extension in general but there are some principles which effectively cause rural development and they play vital role in overall development of people living in rural areas. They are as follows:

Principle of Leadership

Leadership plays crucial role in development of extension system. Local leaders are the custodians of local thought and action. The involvement of local leaders and legitimization by them are essential for the success of a programme. A good rule in extension work is “Never do anything yourself that you can get someone to do for you”. The involvement of leaders in extension programmes is the one single factor that determines the success or failure of any programme. All communities have leaders or potential leaders. They need to be identified and engaged in extension programmes as they can interpret the innovations in convincing way to the village people. Due to trust, a new agricultural practice will be adopted more easily if it is recommended by such farmers. The opinion leaders and information controllers play a critical role in effective in communicating ideas in the rural system.

Principle of Cooperation and Participation

Cooperation is the basis for existence of an extension service. People must participate in the development of a programme and must feel that it is their own programme. The idea behind the extension work is the coming together of the rural people for social upliftment. The extension programmes should be the people’s programme with Government aid. All should cooperate and help each other in this process. Good extension worker helps the rural people in identifying their problems and then helping them in solving these problems. People will not feel attached to the work which is thrust upon them without their consultation.

Principle of Democratic Approach

Extension operates with consensus through discussions and suggestions. The problems and possible solutions need to be discussed with the people. It can bring all possible alternative solutions and they get chance to weigh their

merits and demerits and accordingly adopt a solution through mutual discussion. Ultimately, the people are left free to decide their line of action, the methods to be adopted in the local situation with their own resources and available government assistance.

Principle of Cultural Differences and Cultural Change

Every culture has a distinct way of life, a set of attitudes and values. As each culture is unique and each particular situation within which a change is occurring, or is to be made, is unique, it is not possible to lay down prescription for any course of action. It is desirable to discover and understand the cultural values and rituals as well as perception of people before pushing the agenda of change. We must have. With dialogue and mutual respect and receptiveness for local knowledge, wisdom and values, acceptance of change among the people could be motivated.

Principle of Trained Specialists

Like other sciences, agriculture, animal husbandry and the home science are moving forward rapidly. Therefore, maintaining competency in any one of these fields is a continuous job. It is very difficult for an extension worker to know the latest findings of research in all the branches of science to deal with his day-to-day activity. As stated by Mosher (1958), it is impossible for any field agent to be really an expert in all the problems for which he is called upon to help. To supplement his abilities, and to keep field agents informed, both about new developments and market trends, it has been found necessary to have field agents supported by subject matter specialists. A subject matter extension specialist is responsible for keeping his knowledge up-to-date on research development, for training field extension staff in the latest techniques and recommend practices in his field of specialization.

Principle of Use of Local Resources

We generally observe that a number of local resources, both human and material are not put to their full use. The adult and the youth are usually without work during the slack season. Under these conditions, it is the duty of the extension staff to mobilize and organize the local resources of men and material and the social groups for joint and cooperative action. As far as possible, efforts should be made to plan and execute all programmes locally. It is not very difficult to identify the local resources for executing various programmes. It, then helps to make self-development as the habit of the people which is a pre-requisite for any nation to progress.

Grassroots principle of organization

Any strategy for development of rural people must keep in consideration the local views and aspiration. Bottom-up planning has been favoured to facilitate local representativeness. Therefore, local organizations like Panchayat Raj Institutions, farmers' organizations like cooperatives, FPOs,

SHGs or FIGs must be included in planning and implementation of extension programmes.

Principle of learning by doing

The farmers must be motivated and encouraged for participation in testing a solution themselves. The motive for change must come from them. If they try a new method, they learn better. It will. Besides conviction about the technology, the outcome will build confidence and infuse a sense of experimentation ability and inquisitiveness among them. They can improve a technology through renovation and also developing a new technology.

Principle of whole family approach

Extension should lay focus upon engaging whole family as besides the male, women too contribute significantly in farming. The new knowledge and skills should be accessible to all. There is growing emphasis upon family farming, where-in the production is managed and operated by family relying predominantly on family capital including labour, both men's and women's.

Principle of satisfaction

The extension approach and intervention must be able to generate satisfaction among the after using a new method. If the users are satisfied with new method, it is likely that they will go for repeated use at a larger scale. Their sharing of satisfactory experiences will motivate others to adopt it and a chain of diffusion of the new method will be triggered in the social system.

12.4 EXTENSION APPROACHES

World-wide several forms of extension had been practised. Axinn (1988) classified them into eight major approaches in **Guide on Alternative Extension Approaches**. These are briefly summarized as below:

- 1) **The general agriculture extension approach:** The approach is usually fairly centralized, top-down and government-controlled. The success is measured in the adoption rate of recommendations and increases in national production.
- 2) **The commodity specialized approach:** Extension is oriented towards one commodity or crop and the goals are intentionally production and profit-oriented. All aspects of production and marketing are vertically integrated.
- 3) **The training and visit approach:** This approach was based on a rigorously planned schedule of visits of extension agents to farmers and training of extension agents and subject matter specialists, SMS.
- 4) **The agricultural extension participatory approach:** This approach often focuses on the expressed needs of farmers' groups and its goal is increased production and an improved quality of rural life. Implementation is often decentralized and flexible.

- 5) **The project approach:** This approach concentrates efforts on what is needed and required by the beneficiaries and the donors, meaning that projects to be implemented must be a consensus of the general audience and must be necessary and demand driven as projects identified must meet the immediate needs of the beneficiaries or the end users.
- 6) **The farming systems development approach:** A key characteristic of this type of extension is its systems or holistic approach at the local level. Close ties with research are required and technology for local needs is developed locally through an iterative process involving local people.
- 7) **The cost-sharing approach:** This approach assumes that cost-sharing with local people (who do not have the means to pay the full cost) will promote a programme that is more likely to meet local situations and where extension agents are more accountable to local interests.
- 8) **The educational institution approach:** This approach uses educational institutions which have technical knowledge and some research ability to provide extension services for rural people.

Check Your Progress 2

Note: i) Use the space below for you answers.

ii) Compare your answers with those given at the end of this unit.

- 1) Should India promote cost-sharing approach of extension?

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- 2) What are the limitations of top-down approach of extension?

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12.5 INNOVATIONS AND REFORMS IN EXTENSION IN INDIA

Extension system in India has witnessed several reforms. During pre-Independence era rural extension was predominantly philanthropic. After independence, through the government initiatives began it had an organized structure and functions. Extension efforts and programmes carried out by government institutions is also referred as public extension system. Public

extension system has been the backbone of technology delivery in India. Beginning from Community Development Programme (1952) and National Extension Service (1953), Ministry based extension has remained the core extension system, though the front-line extension system of ICAR/State Agricultural Universities too has played a critical role in technology delivery and capacity building of farmers as well as extension professionals. There is emergence of pluralism in extension with significant presence of multiple players in technology dissemination such as financial institutions, corporate and private firms, input agencies, civil societies, and also farmers' organizations. Reforms in extension system have provided a governance structures, capacity, organization and management, and advisory methods. A brainchild of Benor and promoted by the World bank, the Training & Visit (T&V) system of extension was launched in 1974 in India, which provided a re-organized extension set up in the country with streamlined operational framework. Professionalism, Single line of Command, Concentration of Effort, Time-bound work, Field and Farmer Orientation, Regular and Continuous Training, and Linkage with Research were the salient features of T&V programme. It prescribed a regular training of extension workers and their visits to farmers on fortnightly basis. Contact farmers were identified and trained for imparting knowledge and skills to fellow farmers. Though T&V strengthened the extension system in India, the limitations of finance as well as lack of inclusiveness and top-down approach retarded its effectiveness.

12.5.1 Support to State Extension Programmes for Extension Reforms

The recent reform has emphasized upon bottom-up planning, participatory approach, stakeholders' representativeness, focus on extension education, scope for entrepreneurship development through monetary support and technological backstopping, promotion of farmers' organizations, and market linkage initiatives through implementation of Agricultural Technology Management Agency(ATMA).

The Scheme "Support to State Extension Programmes for Extension Reforms" was pilot tested from 1998 to 2005. The scheme aims at following:

- Encouraging Multi-Agency Extension Strategies involving Public/Private Extension Service Providers.
- Ensuring an Integrated, Broad-Based Extension Delivery Mechanism consistent with Farming System Approach with a focus on Bottom Up Planning process.
- Adopting Group Approach to Extension in line with the identified needs and requirements of the farmers in the form of CIGs & FIGs.
- Facilitating Convergence of Farmer Centric Programmes in planning, execution and implementation.

- Addressing Gender Concerns by mobilizing farm women into groups and providing training to them.

ATMA was introduced in 1998 to introduce reforms in the public sector agriculture extension system and increase its relevance and accessibility among all the stakeholders under Innovation in Technology Dissemination (ITD) component of World Bank funded National Agricultural Technology Project (NATP). ATMA was launched with a view to coordinating all on-going extension efforts in the district and facilitate a platform for convergence of efforts and resources of government, civil society, farm community and private sector. A platform of convergence, joint programming, fund flow and extended outreach to farmers are the major benefits perceived by the officials of agriculture and allied departments. Bottom-up planning, inclusions of farmer's voice and choice, planning of need based interventions and spurt to extension activities of line departments with fund flow and grassroots network have been the silent benefits of ATMA. With creation of Farmer Advisory Committees at block and district levels, bottom up planning was promoted. It helped in engaging the progressive and innovative farmers in planning, diffusion of innovations and capacity building. Farm schools were promoted to facilitate lateral sharing and diffusion of innovations among the farmers through innovative farmers and experts.

ATMA is a registered society, an autonomous institution, at the district level. ATMA governing board (AGB) is the apex body chaired by the district magistrate (Fig.1), while The ATMA Project Director chairs the ATMA Management Committee (AMC), which coordinates the implementation of extension activities at district level. At the block level there is Block Technology Team (BTT) and Farmer Advisory Committee (FAC) which prepare Block Action Plan.

At the state-level, an Interdepartmental Working Group (IDWG) formulates a State Extension Work Plan (SEWP) based upon the Strategic Research and Extension Plan (SREP) of the districts. The SREP and SEWP are the instruments that promote convergence of extension activities between line departments and research institutions at the district and state levels, respectively. The State Nodal Cell (SNC), consisting of State Nodal Officer, State Coordinator, State Gender Coordinator and Supporting Staff formulates the State Extension Work Plan (SEWP) and the State Level Sanctioning Committee (SLSC) set up under Rashtriya Krishi Vikas Yojana (RKVY) is the apex body which approves State Extension Work Plan (SEWP).

A State Agricultural Management and Extension Training Institute (SAMETI) has been created in every state to take activities of capacity building of extension functionaries in the concepts and processes of ATMA.

At village level, "Farmer Friend" has been designated in order to bridge the gap between farmers and extension agents. One Farmer Friend caters to two

villages and helps in mobilizing farmers and assists in organizing extension activities.

National Mission on Agriculture Extension and Technology (NMAET)

NMAET was introduced in 2014-15 and it aimed to restructure and strengthen agricultural extension to enable delivery of appropriate technology and improved agronomic practices to farmers. This is envisaged to be achieved by a judicious mix of extensive physical outreach and interactive methods of information dissemination, use of ICT, popularisation of modern and appropriate technologies, capacity building and institution strengthening to promote mechanisation, availability of quality seeds, plant protection etc. and encourage aggregation of Farmers into Interest Groups (FIGs) to form Farmer Producer Organisations (FPOs).

It had four sub-missions viz, (i) Sub Mission on Agricultural Extension (SMAE), (ii) Sub-Mission on Seed and Planting Material (SMSP), (iii) Sub Mission on Agricultural Mechanization (SMAM) and (iv) Sub Mission on Plant Protection and Plant Quarantine (SMPP).

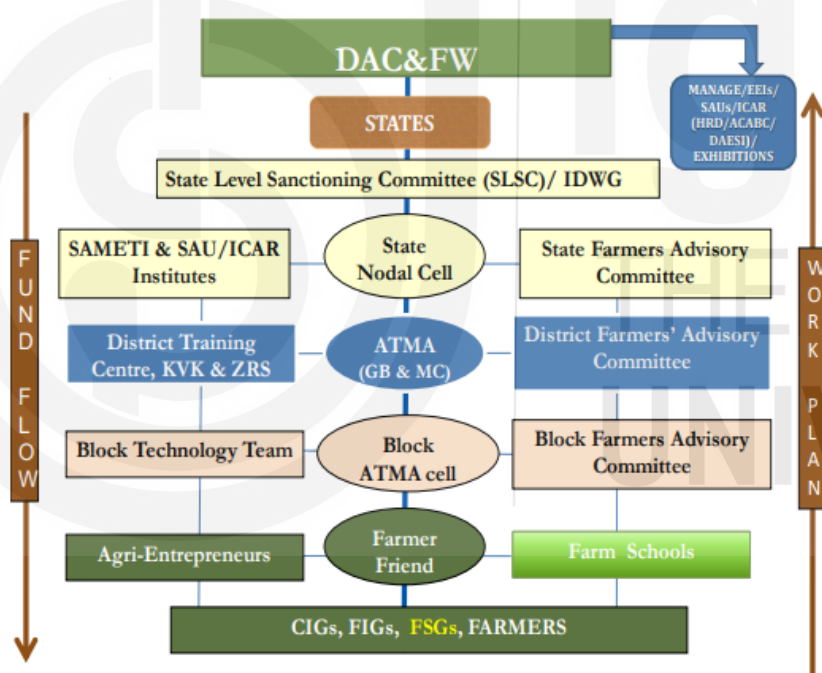


Fig.1: Organizational Structure of ATMA; Source: ATMA GUIDELINES, 2018 under Krishonnati Yojana

12.5.2 Extension Through Educational Institutions

The Indian Council of Agricultural Research (ICAR) has spearheaded frontline extension system in India through National Demonstration Programme (1965), Operational Research Programmes (1974-75), Farm Science Centers/Krishi Vigyan Kendras (KVKs) (1974) and Lab to Land Programme (1979).

National Demonstration Programme was launched to demonstrate the production potentialities of technology package on major crops. It played a

pivotal role in ushering Green Revolution. **Operational Research Project** aimed at disseminating the proven technology in a discipline/area among farmers on a watershed basis covering the whole village or a cluster of villages and concurrently studying constraints (technological, extension or administrative) as barriers to the rapid spread of improved technical know-how. The first **KVK** was established in 1974 at Pondicherry under Tamil Nadu Agricultural University under the recommendations of Mohan Singh Mehta Committee. KVK's are designed to impart need-based and skill-oriented vocational training to the practicing farmers, in-service field level extension workers, and to those who wish to go in for self-employment. **Lab-to-Land Project** was launched by the ICAR in 1979 as a part of its Golden Jubilee celebration. The overall objective of the programme was to improve the economic condition of the small and marginal farmers and landless agricultural labourers, particularly scheduled castes and scheduled tribes, by transfer of improved technology developed by the agricultural universities, research institutes etc. With effect from **April 1992**, all first-line transfer of technology projects of the ICAR viz. ND, ORP and LLP were integrated into KVKs.

A KVK is a multidisciplinary educational institution situated at the district level, with funding and technical supervision from ICAR and with 721 KVKs it forms the largest network of frontline extension system in the country. The major activities include (i) On-farm testing of agricultural technologies to assess their location specificity of under various farming systems (ii) Organize Frontline Demonstrations to establish production potential of technologies on the farmers' fields, (iii) Capacity development of farmers and extension personnel to update their knowledge and skills on modern agricultural technologies, (iv) To work as knowledge and resource centre of agricultural technologies for supporting initiatives of public, private and voluntary sector in improving the agricultural economy of the district, (v) Provide farm advisories using ICT and other media. It forges linkages and coordination with state department of agriculture for technology backstopping and capacity building.

12.5.3 New Initiatives of ICAR

Farmer FIRST, Attracting and Retaining Rural Youth in Agriculture (ARYA), **My Village My Pride/Mera Gaon Mera Gaurav (MGMG)** and Knowledge Systems and Homestead Agriculture Management in Tribal Areas (**KSHAMTA**) are the new initiatives of ICAR for the technology transfer and capacity building for enhancing productivity and income of farmers and youth.

- a) **Farmer FIRST (Farm, Innovations, Resources, Science and Technology):** It has been initiated since October, 2016. The farmer is in a centric role for research problem identification, prioritization and conduct of experiments and its management in farmers' conditions. It focuses upon farmer's farm, innovations, resources, science and

technology (FIRST) to provide impetus to development of small farmers operating in complex, diverse and risk-prone situations. It aims at enriching Farmers–Scientist interface, technology assemblage, application and feedback, partnership and institutional building and content mobilization.

- b) **ARYA** was launched on 25th July, 2015, which aims at the increasing participation of youth, to sustain agriculture and allied activities through scientific methods.
- c) **Mera Gaon Mera Gaurav (My Village My Pride)** was launched on 17th October, 2015 and aims at strengthening farmers-scientists interface. A multidisciplinary team of 4 scientists work in a cluster of 5 villages for agricultural development.
- d) **Student READY (Rural Entrepreneurship Awareness Development Yojana)**: Launched by Hon'ble Prime Minister on 25 July 2015, the programme was conceptualized to reorient graduates of Agriculture and allied subjects for ensuring and assuring employability and to develop entrepreneurs for emerging knowledge intensive agriculture by articulating knowledge, skill, ability and experiences.
- e) **Knowledge Systems and Homestead Agriculture Management in Tribal Areas (KSHAMTA)**: KSHAMATA is launched with the objective of achieving balanced growth and economic development in the country with special focus of tribal agriculture and in line with the mission statement of Government of India “*Sabka Sath Sabka Vikas*”.
- f) **Nutri-sensitive Agricultural Research and Innovations (NARI)**: The programme aims to promote nutrition-sensitive agriculture, capacity development and gender mainstreaming
- g) **Value Addition and Technology Incubation Centers in Agriculture (VATICA)**: It was conceptualized by ICAR to create a facility to provide incubation training to rural youth in processing and value addition.

The SAUs are another important arm for promoting extension activities in the states. While their main mandate is formal degree programmes in major agricultural disciplines, they provide extension and training support through the directorate of extension and education. The extension plan is developed through Extension Council and the packages of practices are developed based upon the findings of regional stations and KVKs. The final recommendations are disseminated through KVKs, project activities ICT channels and media.

Innovative Extension Approaches

Digital extension approach

Extension reforms brought a transformation in agricultural extension system through introduction of Information and Communication Technologies (ICTs). The ICT-enabled extension system referred to as Digital Extension

has the potential for enabling the empowerment of farming communities by improving their access to information and sharing knowledge with innovative e-agriculture initiatives (Saravanan, 2010). Farmers Portal (a single platform where farmers can access information on all agriculture related issue), mKisan (an SMS portal which gives information in the local language); Kisan Call Centre (toll-free, phone based agricultural advisory services in local languages); and DD Kisan (a television channel on agriculture); community radios are the major ICT initiatives of the government. Warana wired village project; ITC's e-Chaupal, village kiosk based extension of Drishtee; video based extension of Digital Green; Pusa mKRISHI mobile advisory; video based Pusa Samachar; Gramdoot, Gyandoot, Tarahaat, Akshaya, etc. have been successful ICT enabled extension.

Extension through private sector

There is a growing presence of private sectors in making interventions of organizing exhibitions and awareness camps on climate change preparedness, nutrition, conservation of resources, use of clean energy, employment generation, technology demonstrations, and capacity building for agricultural development and rural upliftment. Some cases are listed below:

Mahindra Krishi Vihar: Mahindra and Mahindra Limited has established one-stop solution centres with the establishment of Mahindra Shubhlabh Services subsidiary. The one-stop solution centres operate on a franchise basis and provide quality inputs, rent farm equipment, credit in partnership with banks, farm advice by trained field supervisors who visit fields, and arrange contracts with processors for off-take of crop produce.

Tata Kisan Sansar by Tata Chemicals Ltd: It is a franchise-based 'hub and spokes' model of outlets. The extension services offered by these one-stop shops include soil testing, remote diagnostics and house brands for seeds, cattle feed, pesticides and sprayers.

Godrej Agrovet has chain or rural outlet and run in partnership with other companies to extend its products range. Its 'one-stop solutions' model offers agricultural equipment, consumer goods, technical services, soil and water testing, veterinary, financial and post office services, and pharmaceuticals.

12.5.4 Extension through NGO

Civil organizations or non-government organizations (NGOs) have a very strong connect with rural people. Organizations BAIF, Basix and PRADAN operate in numerous states and have been active in extension services for many years.

The Bharatia Agro-Industries Foundation (BAIF): BAIF is a Gandhian organization committed to impart livelihood opportunities to the rural underprivileged families. BAIF's areas of work include water resources development, sericulture, agro forestry, post-harvest product management

and marketing cattle feed and forage production. BAIF has facilitated the formation of farmers' cooperatives and federations of self-help groups.

Basix: Basix strategy is to provide a comprehensive set of livelihood promotion services which include Financial Inclusion Services (FINS), agricultural/Business Development Services (Ag/BDS) and institutional Development Services (IDS) to rural poor households under one umbrella. It provides services related to productivity enhancement, risk mitigation, local value addition and alternative market linkage for synthetic bio-inputs and outputs.

PRADAN: A PRADAN works in families which belong to the Schedule Tribes and Schedule Castes. PRADAN promotes the Integrated Natural Resources Management (INRM) of land, water, forest and biological resources to achieve and sustain potential agricultural productivity.

Public-Private Partnership Based Extension

The term public private partnership refers to a collaborative effort of public and private sectors to accomplish a shared goal. PPP based extension aims at making a joint efforts of public and private extension agencies for providing quality advisory, capacity building and market linkages. Public-Private Partnership based extension is an effective means to reduce the financial burden of extension service on public extension system and to meet the emerging advisory needs related to agri-business and market-orientation. ATMA has facilitated such partnerships. Partnership of Department of Agriculture, Madhya Pradesh and Dhanuka Pvt. Ltd. is one the pioneering efforts to work together in areas like soil testing, training, farmers tour programmes, demonstrations, transfer of technology. Thirumadhuram Pineapple project involving Kudumbhasree Project Mission, Department of Agriculture, women SHGs and Nadukkora Agro processing centre (Rajendran et al., 2010) is a successful PPP model.

IARI-VO partnership model: IARI-Voluntary Organizations Partnership Extension Model (IVOPEM) has been started to assess and disseminate IARI technologies in their operational areas. Location specific technologies were identified and promoted through technology assessment in partnership with the partner VOs. Feedback of the farmers and other stakeholders were provided to the researchers for refinement to make the technologies location specific and problem solving. Two (2) VOs (Young Farmers Association, Rakhra, Patiala and PRDF, Gorakhpur) started seed production of improved IARI varieties of paddy and wheat and became a seed hub for regional seed self-sufficiency.

Check Your Progress 3

Note: i) Use the space below for you answers.

ii) Compare your answers with those given at the end of this unit.

1) What are the advantages and disadvantages of private extension system?

- 2) What are the challenges of partnership based extension?

12.6 LET US SUM UP

This unit has tried to introduce the concept of rural extension with respect to rural development. Various concepts such as development, extension, rural extension, rural development have been covered. Critical elements, different paradigms and growth of different rural developmental programmes have been discussed under various sections. Despite all this developmental initiatives, the gap between technology developed/released and the technology disseminated/adopted is wide. The greatest challenge with extension is how to transform and strengthen pluralistic agricultural extension and advisory systems in moving toward the broader goal of increasing farm income and improving rural livelihoods. This can be achieved by improving extension management, decentralization, single commodity focus, fee for service public provision, institutional pluralism, empowerment, participatory approaches, privatization, interconnecting rural people and the use of appropriate media. Towards the end of the unit, we have discussed some examples of extension services of different agencies.

12.7 KEY WORDS

Rural Extension is an educational process which works with rural people, supports them and prepares them to confront their problems with an ultimate aim of rural development.

Rural development is a process leading to sustainable improvement in the quality of life of rural people, especially the poor.

Development is a process that creates growth, progress, positive change or the addition of physical, economic, environmental, social and demographic components.

Innovation is described as a horizontal process in which practices are changed as a result of social experiences and takes into account previous experience of agents.

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